MIL-E-63399B(AR)
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SUPERSEDING
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23 July 1980

MILITARY SPECIFICATION

ELECTRONICS-MCD ASSEMBLY FOR MINE, ANTITANK: HE, M75

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification provides quality assurance provisions for the fabrication, inspection and packing of an electronic assembly used in the Mine, Antitank: HE, M75.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19500/127	-	Semiconductor Device, Diode, Silicon, Voltage-Regulator Types IN4370A through IN4372A, IN746A through IN759A, IN4307A-1 through IN4372A-1, IN746A-1 through IN759A-1, JAN, JANTX, JANTXV, and JANS.

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-19500/240C	- Semiconductor Device, Diode, Silicon, Rectifier, Types IN645, IN647, IN649, IN645-1, IN647-1, IN649-1, TX and Non-TX Types.
MIL-S-19500/255	- Semiconductor Device, Transistor, NPN, Silicon, Switching Types 2N2221, 2N2221A, 2N2222, and 2N2222A, JAN, JANTX, JANTXV, and JANS.
MIL-S-19500/291	- Semiconductor Devices, Transistor, PNP, Silicon Switching Types 2N2906, 2N2906A, 2N2907, and 2907A, JAN, JANTX, JANTXV, and JANS.
MIL-R-39008	- Resistors, Fixed, Composition, (Insulated).
MIL-R-39008/1	- Resistor, Fixed, Composition (Insulated) Established Reliability Style RCR07.
MIL-C-39014/5-2325	- Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability, Style CKR12.
MIL-S-45743	- Soldering, Manual Type, High Reliability Electrical and Electronic Equipment.
MIL-S-46844	- Solder Bath Soldering of Printed Wiring Assemblies.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-P-50884	- Printed Wiring, Flexible.
MIL-P-55110	- Printed Wiring Boards Specification For.
MIL-A-63242	- Amplifier, Operational, Triple Programmable Micropower

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MIL-P-63316 - Primer, Battery, Electric Parts for and Loading, Assembling and Packing.

MIL-M-63320 - Microcircuit, Digital, CMOS, (Initial Logic).

MIL-M-63321 - Microcircuit, Digital, CMOS, (Final Logic).

MIL-N-63396 - Network, Resistor - Fixed and Adjustable.

MIL-N-63400 - Network, Capacitance - Resistor.

MIL-P-63421 - Power Supply Assemblies for Fascam Electronics Assemblies.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-202 - Test Methods For Electronic and Electrical Component Parts.

MIL-STD-275 - Printed Wiring for Electronic Equipment.

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for.

MIL-STD-750 - Test Methods For Semiconductor Devices.

MIL-STD-883 - Test Methods For Microelectronics.

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DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

7258943 - Shipping and Storage Container, M548.

9287115 - Potting Material.

9287148 - Encapsulation Procedure and Requirements.

9292196 (Sec) - M75 Electronics Assembly Performance Requirements.

9292225 - Electronics-MCD Assembly.

9292220 - Capacitors, Tantalum, Dry Electrolyte, Polarized, Reliability Requirements For.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2240-68 - Indentation Hardness of Rubber and Plastic by Means of a Durometer, Test for,

ASTM D412-68 - Tension Testing of Vulcanized Rubber

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawings (DWGS) 9292225 and 9292196 and associated drawings, and with requirements specified in applicable specifications and standards.

3.3 Physical characteristics.

3.3.1 Envelope and dimensions. The envelope and dimensions of the Electronics-MCD Assembly shall conform to the requirements on DWG 9292225, as applicable.

3.3.2 Potting. Potting material shall comply with the requirements of DWGS 9287115 and 9287148, and shall be manufactured and dispensed in accordance with the processes, procedures, and controls delineated therein. The potting material shall be tested as specified in 4.4.3.2 and 4.5.27. The potted Electronics Assembly shall not show evidence of voids, bubbles and delaminations exceeding the following criteria:

a. Bubbles within the sprue vent area are acceptable provided they do not contact the top or bottom surface of the printed wiring board (PWB). A bubble adjacent to a diode or metal film resistor may not contact the component for a length greater than the diameter of that component. Bubbles in the remaining assembly are acceptable, provided they are less than one-quarter (1/4) inch in their maximum physical dimension and do not contact 10% or more of the surface area of any electrical component.

b. Bubbles exposing any position of the lead termination or junction are allowed provided there are no more than four (4) per unit and the exposed area is less than 25% of the diameter of the component.

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c. Delaminations inward from the junction of the housing, lid and PWB shall not exceed .200 inches. The cumulative length of all such delaminations shall not exceed more than 20% of the circumference of the board. The total cumulative area of other delaminations shall not exceed one (1) square inch.

d. Sprue cutoff shall be flush to .04 inches below the surface. The potted Electronics Assembly shall be inspected using ARRADCOM approved visual inspection aids.

3.4 Environmental conditions

3.4.1 Operating temperature. When connected to an external d.c. power supply, the Electronics-MCD Assembly (electronic assembly) shall activate and perform as specified in DWG 9292196 over the temperature range given therein.

3.4.2 Transportation vibration. The electronic assemblies packed for shipment shall withstand and show no evidence of internal or external damage that will affect the intended functions when subjected to transportation vibration specified in 4.5.25.

3.5 Electrical requirements

3.5.1 Self-destruct, timing control and firing circuit characteristics. The electronics assembly shall provide timing controls and firing signals as specified in DWG 9292196 and the following paragraphs. Unless otherwise specified, V_B = 6.75 volts and ambient temperature (T_A) shall be 25°C.

3.5.2 Quiescent current. Electronic assembly quiescent current shall be less than 288 microamperes when the supply voltage is 6.75 ± 0.15 volts, the ambient temperature 25°C ± 5°C and the input/output terminals loaded as specified in 4.5.1.

3.5.3 Timing, signal, and detection operations

3.5.3.1 Primary Time Base (PTB) oscillator period. The PTB oscillator period shall comply with 3.4.1 when tested as specified in 4.5.2.

3.5.3.2 Test Time Base (TTB) oscillator period. The TTB oscillator period shall comply with 3.4.1 when tested as specified in 4.5.3.

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3.5.3.3 Arming actuator signal. The PTB clock count at which the arming actuator signal turns on the piston actuator firing circuit shall comply with 3.4.1 when tested as specified in 4.5.4.

3.5.3.4 Sensor enable. The PTB clock count at which the electronic assembly is sensor enable shall comply with paragraph 3.4.1. The electronic assembly shall not process destruct signals from the low voltage detect, prime target sensor and antidisturbance circuits until after sensor enable. The PTB clock count that initiates sensor enable and the ability of the electronic assembly to inhibit destruct signals prior to sensor enable shall be tested as specified in 4.5.5.

3.5.3.5 Low voltage detect. The supply voltage level at which the low voltage detect circuit turns on the detonator firing circuit shall comply with 3.4.1 when tested as specified in 4.5.6.

3.5.3.6 Extended self-destruct signal. The PTB clock count at which the extended self-destruct signal turns on the detonator firing circuit shall comply with 3.4.1 when tested as specified in 4.5.7.

3.5.3.7 Normal self-destruct signal. The PTB clock count at which the normal self-destruct signal turns on the detonator firing circuit shall comply with 3.4.1 when tested as specified in 4.5.8.

3.5.3.8 Target sensor functioning. The electronic assembly shall provide a fire signal (3.5.3.8.b) when subjected to valid target signatures. The specific conditions and signal limits under which this function shall occur are specified in DWG 9292196 and the following paragraphs.

a. Spurious output signal. There shall be no output occurring at the magnetic drive point (Test Point 7) for a minimum of ten (10) seconds prior to enable after initial application of V_B, when tested as specified in 4.5.9.

b. Sensor output voltage. The output from the target sensor (Test Point 1) shall be a minimum of 2.5 volts when the assembly is subjected to a valid signature as specified in DWG 9292196 and when tested as specified in 4.5.9.

3.5.3.9 Antidisturbance mode. The antidisturbance circuit shall initiate the turn on of the detonator firing circuit when the antidisturbance switch changes state; that is, a transition from either an open or closed switch condition. This requirement shall be tested as specified in 4.5.10.

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3.5.3.10 Primary Time Base (PTB) oscillator circuit check. The oscillator check circuit shall initiate the turn on of the detonator firing circuit when the primary time base oscillator period becomes greater than the test time base oscillator period. This requirement shall be tested as specified in 4.5.11 and comply with 3.4.1.

3.5.3.11 Detonator fire signal output energy. The electronic assembly shall provide energy to fire the piston actuator or the parallel detonators when required as described in this specification. The output energy shall be such that a minimum of three (3) volts shall be measured no more than forty (40) microseconds after a fire output signal is generated, and a minimum of two (2) volts is measured at a minimum time of 120 microseconds after the fire output signal is generated. Voltages shall be measured at the output terminals across a 3.9 ohm ± 5% resistive load.

a. Piston actuator firing circuit. When turned on, the piston acutator firing circuit shall remain on for only one PTB clock count, that is, a time interval equal to one PTB oscillator period when tested as specified in 4.5.12.

b. Detonator firing circuit. When turned on, the parallel detonators firing circuit shall remain on for as long as the electronic assembly is functionally operational or it is intentionally reset when tested as specified in 4.5.12.

3.5.3.12 Self-sterilization. Within 5 seconds after the turn on of the detonator firing circuit the voltage on capacitor C12 shall be less than 1.3 volts. It shall remain less than 1.3 volts for as long as the detonator firing circuit remains on. This requirement shall be tested as specified in 4.5.13.

3.5.3.13 Reset circuit. When a linear increasing voltage (zero to 6.75 ± 0.15 volts in 50 milliseconds min) is applied to the electronic assembly from an external d.c. power supply, the reset circuit (C11, R7) shall reset the test points A & B to a logic low. This requirement shall be tested as specified in 4.5.14.

3.5.4 Magnetic Coupling Device (MCD) Assembly Resistance. The resistance across the MCD coil terminals when installed in the Electronics Assembly shall be as specified on DWG 9292225 when tested as specified in 4.5.15.

3.5.5 Shorting Bar Assembly continuity. The resistance across each shorting bar when installed in the electronic assembly shall be as specified on DWG 9292225 when tested as specified in 4.5.16.

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3.5.6 Non-activated voltage. Each battery cell shall be unactivated and measure less than .005 volts when tested as specified in 4.5.17.

3.5.7 Soldering. All soldering electrical connections shall comply with MIL-S-45743 and MIL-S-46844, except as specified in Appendix herein.

3.5.7.1 Solderability. Electronic and electric assembly parts that are soldered shall comply with the solderability requirements of MIL-S-45743 and MIL-S-46844, except as noted otherwise.

3.5.7.2 Quality assurance soldering plan. The contractor will generate a Quality Assurance Soldering Plan that will detail the overall soldering procedure. This plan shall include detailed information for handling, cleaning, hand soldering, machine soldering, touch up or rework, process controls, tool maintenance, flagging procedures for avoiding excessive defects and all other pertinent QA procedures relative to soldering. The plan shall include the documentation requirements of MIL-S-46844. This plan shall be submitted to AMRADCOM, DRDAR-QAT-M (see 4.5.20, 6.6).

3.6 Inspection data. Data shall be recorded and retained for all first article and quality conformance tests required by this specification in accordance with 4.5.21.

3.6.1 Variables data. The Primary Time Base Oscillator Period, Test Time Base Oscillator Period, Quiescent Current, Low Voltage Detect Circuit Trigger Level, and the high and low Magnetic Sensor Threshold Levels shall be measured and recorded as variable data for first article samples and surveillance units (see 4.5.22).

3.7 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Surveillance units. The contractor shall provide for surveillance quantities of Electronic-MCD Assemblies from each month's production as specified in the contract (see 4.5.23, 6.2).

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3.9 Workmanship. The electronic assembly shall be fabricated and finished in a thorough, workmanlike manner. It shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust corrosion productions, and other foreign matter. The cleaning method used shall not be injurious to any components, nor shall they be contaminated by the cleaning agent. All markings and stampings, shall be neat and sharply defined. The electronic assembly shall be manufactured or processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The Contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

PART DESCRIPTION	DRAWINGS	QUANTITY
Magnetometer Assembly, (prior to potting)	9287146	10
Shorting Bar and Connector Assembly	9292158	10
Printed Wiring Board Assembly	9292223-2	10
Electronics Assembly	9292224-2	18
Electronics-MCD Assembly	9292225-1	16
Electronics-MCD Assembly	9292225-2	5
MCD Assembly	9292231	10
Lid, Electric Assembly	9292243	10
Housing, Electric Assembly	9332372	10
Printed Wiring Board	9345328	10

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4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Requalification. The electronic assembly shall be subjected to another first article as stipulated in the contract if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function of the assembly are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocated his assembly facilities.

c. The contractor uses a new sub-contractor for material, parts, or assembly.

d. A change occurs in design, material, performance or assembly.

e. A lapse in production for a period exceeding 90 days or as stipulated in the contract.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Electronics-MCD Assy and Sub-Assemblies				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Printed Wire Board (DWG 9345328)	10			
	Examination for defects		--	3.2	4.4.2.1
	Solderability			3.2	4.4.2.1
	Printed Wiring Board Assy (DWG 9292223-2)	10			
	Examination for defects		--	3.2	4.4.2.2
	Electronic Assembly (DWG 9292224-2)	18			
	Examination for defects		--	3.2	4.4.2.3
	Transportation Vibration		--	3.4.2	4.5.25
	Performance Tests - See 1/ (Amb., Hot, Cold)		--	3.5,3.6.	4.4.2.4, 4.5.24
	Electronics - MCD Assy (DWG 9292225-1)	16			
	Examination for defects	all	--	3.2	4.4.2.6
	Continuity/Resistance Tests	all	--	3.5.4,3.5.5	4.5.15, 4.5.16
	Unactivated Battery Voltage	all	--	3.5.6	4.5.17
	SD Normal-Real Time-Destructive (Hot & Cold)	8	--	3.5.3.7	4.4.2.5, 4.5.26
(Info Only)	SD Extended-Real Time-Destructive (Hot & Cold)	8	--	3.5.3.6	4.4.2.5, 4.5.26
	Electronics - MCD Assy (DWG 9292225-2)	5			
	Examinations for defects		--	3.2	4.4.2.6
	Continuity/Resistance Tests		--	3.5.4,3.5.5	4.5.15, 4.5.16
	Unactivated Battery Voltage		--	3.5.6	4.5.17

NOTES: 1/ Critical 1, Specials a and c and Majors 102 and 107 of 4.4.2.4 shall be recorded as variable data.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE: Electronics-MCD Assy and Sub-Assemblies		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Magnetometer Assembly (DWG 9292146)	10			
	Examination for defects		—	3.2	4.4.2.39
	Resistance test		—	3.2	DWG. 9287149
	Sensitivity tests		—	3.2	DWG. 9287149
	Linearity test		—	3.2	DWG. 9287149
	Impedance test		—	3.2	DWG. 9287149
	MCD Assembly (DWG. 9292231)	10			
	Examination for defects		—	3.2	4.4.2.38
	Housing, Electronic Assembly (DWG. 9332372)	10			
	Examination for defects		—	3.2	4.4.2.41
	Lid, Electric Assembly (DWG. 9292243)	10			
	Examination for defects		—	3.2	4.4.2.42
	Shorting Bar and Connector Assembly (DWG. 9292158)	10			
	Examination for defects		—	3.2	4.4.2.46

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of electronic assemblies and sub-assemblies shall contain:

a. Batteries and/or power supply assemblies bearing the same lot interfix number from one manufacturer.

b. Triple operational amplifiers bearing the same lot interfix number from one manufacturer.

c. Integrated circuit, CMOS, target sensors bearing the same lot interfix number from one manufacturer.

d. Integrated circuit, CMOS, timers bearing the same lot interfix number from one manufacturer.

e. Networks, capacitor-resistor, or resistor bearing the same lot interfix number from one manufacturer.

f. Electrical battery primers bearing the same lot interfix number from one manufacturer.

g. Fuses bearing the same lot interfix number from one manufacturer.

h. Electronic assemblies of one part number only.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Forms, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Normal Inspection Level II.

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Quality conformance inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63999B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Board, Printed Wiring		9345328
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	1/	2/		
Major					
101	O.D. of board			3.2	Gage
102	Center hole diameter			3.2	Gage
103	Diameter of "MCD-S6A" terminal holes, 7 places			3.2	Gage
104	Diameter of smallest hole, 41 places			3.2	Master template-Gage
105	Diameter of second smallest hole, 38 places			3.2	Master template-Gage
106	Diameter of third smallest hole, 38 places			3.2	Master template-Gage
107	Diameter of fourth smallest hole, 143 places			3.2	Master template-Gage
108	Diameter of fifth smallest hole, 4 places			3.2	Master template-Gage
109	Diameter of sixth smallest hole, 2 places			3.2	Master template-Gage
110	Any discontinuous front circuit			3.2	Tester
111	Any discontinuous rear circuit			3.2	Tester
112	Solderability			3.5.7.1	4.5.18, 4.5.20
113	Plated through holes integrity			3.2	Visual
114	Visual and dimensional examination			3.2	MIL-P-55110, 3/
Minor					
201	Contamination on board, 2 sides			3.2	Visual, 4.5.18
202	Evidence of poor workmanship			3.9	Visual

NOTES:
1/ 10 randomly selected boards sample.
2/ Reject for one or more defective.
3/ Methods of Examination and Test, MIL-P-55110.

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Quality conformance inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Printed Wiring Board Assembly	1 OF 1	9292223-1/-2
			NEXT HIGHER ASSEMBLY 9292224-1/-2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	One or more fuse open		100%	3.2	Tester
2	Blocking or zener diode has improper polarity orientation; breakover or blocking voltage; open/short circuit (SD circuit)		100%	3.2	Tester
Major					
101	Any component damaged, missing or improperly assembled	1/	0.40%	3.2	Visual
102	Incorrect orientation of the capacitor(s) polarity mark(s)	1/	0.40%	3.2	Visual
103	Switch assy height from board (9292223-2 only)		0.40%	3.2	Gage
104	Height from board to top of fuse, 2 places		0.40%	3.2	Gage
105	Battery lead weld(s) broken, 4 places		2/	3.2	Visual (5X)
106	Solder workmanship		3/	3.5.7	4.5.18, 4.5.20, APPENDIX
107	Conformal coating missing from noted components		0.40%	3.2	Visual (ultraviolet light)
108	Shorting bar improperly nested, 2 places		0.40%	3.2	Visual
109	Shorting bar support foam missing		0.40%	3.2	Visual
110	Air-gap present between support foam and cavity		0.40%	3.2	Visual
Minor					
201	Height from board to top of battery		0.65%	3.2	Gage
202	Height of leads/solder on bottom of board		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Before soldering. Other inspections refers to post soldering. 2/ MIL-STD-105, level II, reject for one or more defectives. 3/ Accept/reject per section 50 of the Appendix. 4/ Handling procedure (see 4.5.19).

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Quality conformance inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Electronics Assembly		9292224-1/-2
			NEXT HIGHER ASSEMBLY 9292225-1/-2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Identification number or color of assembly is incorrect or illegible	1/	100%	3.2	Visual
Major					
101	Depth of "D" hole		0.40%	3.2	Gage
102	Excessive voids/bubbles in potting		0.40%	3.3.2	Visual
103	Potting material characteristics		2/	3.3.2	4.4.3.2, 4.5.27
104	Any surface noted in Note 2 of DWG not free of potting material		0.40%	3.2	Visual
105	Power supply bushing above surface "A" of DWG		0.40%	3.2	Visual
106	Potting at fill and vent sprues above surface "B" of DWG		0.40%	3.2	Visual
107	Terminal pins and pads are not clean		0.40%	3.2	Visual
108	Overall height, max.		0.40%	3.2	Gage
109	Improper mating of electronic lid and housing	1/	0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Inspection is to be performed just prior to potting operation. 2/ Sample in accordance with 4.4.3.2. Potting process shall be verified daily. 3/ Handling procedure (see 4.5.19).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Electronics Assembly (Non-destructive Tests)	1 OF 2	9292224-1/-2
			NEXT HIGHER ASSEMBLY 9292225-1/-2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primary Time Base (PTB) oscilltor period, under min. or over max.		100%	3.5.3.1	4.5.2
2	PTB clock count at extended self-destruct signal initiation, over max.		100%	3.5.3.6	4.5.7
3	PTB clock count at normal self-destruct signal initiation, over max.		100%	3.5.3.7	4.5.8
4	PTB clock count at S&A assembly arm signal initiation, under min.		100%	3.5.3.3	4.5.4
Special*	(* see 6.7)				
a	Test Time Base (TTB) oscillator period, over max.		100%	3.5.3.2	4.5.3
b	Oscillator check circuit does not detect a PTB period that is greater than the TTB period		100%	3.5.3.10	4.5.11
c	Low voltage detect circuit trigger level incorrect		100%	3.5.3.5	4.5.6
d	Self-sterilization circuit function incorrect		100%	3.5.3.12	4.5.13
Major					
101	PTB clock count at S&A assy arm signal over max.		4.4.3.1	3.5.3.3	4.5.4
102	Quiescent current over max.		4.4.3.1	3.5.2	4.5.1
103	PTB clock count at sensor enable, incorrect		4.4.3.1	3.5.3.4	4.5.5

NOTES:
1. All units shall be tested, at ambient, for functional parameters listed under critical and special.
2. Except as noted in 4.4.3.1, a sample of the units shall be tested for all the functional parameters listed under critical, special and major at ambient, hot and cold.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4 Cont'd	Electronics Assembly (Non-destructive Tests)				9292224-1/-2
					NEXT HIGHER ASSEMBLY 9292225-1/-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
104	Sensor enable test - low voltage detect, magnetic sensor, and antidisturbance destruct signals not inhibited before sensor enable (AD applies to 929224-2 only)		4.4.3.1	3.5.3.4	4.5.5
105	TTB oscillator period under min.		4.4.3.1	3.5.3.2	4.5.3
106	Antidisturbance switch mode function incorrect (9292224-2 only)		4.4.3.1	3.5.3.9	4.5.10
107	Target sensor function test		4.4.3.1	3.5.3.8	4.5.9
108	Piston actuator firing circuit energy output incorrect		4.4.3.1	3.5.3.11a	4.5.12
109	Detonator firing circuit energy output incorrect		4.4.3.1	3.5.3.11b	4.5.12
110	PTB clock count at extended self-destruct signal, under min.		4.4.3.1	3.5.3.6	4.5.7
111	PTB clock count at normal self-destruct signal, under min.		4.4.3.1	3.5.3.7	4.5.8
112	Reset circuit function incorrect		4.4.3.1	3.5.3.13	4.5.14
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
3. Report all critical and special defects per 4.5.21.
4. Handling procedures (see 4.5.19).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Electronics - MCD Assembly Destructive Tests (Hot and Cold)	1 OF 1	9292225-1/-2
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Performance: INITIATE BATTERIES AND FUNCTION IN "REAL TIME" IN SD MODE.				
Critical					
1	Premature arming signal	1/	0-1		
2	Premature firing	2/	0-1		
3	Hazardous dud	3/	0-1		
Major					
101	Self-destruct (SD) Normal/Extended	*	—		4.5.26
102	Dud		—		
103	Early SD	4/	—		

SD Extended: For information only

*	Inspection Lot Size	Sample Size 5/ SD(Normal)	Sample Size 5/ SD(Extended)	Accept - Reject (For SD Normal) Early	Hazardous ---Dud---	Premature Arming/Firing	Duds
	501-1200	6	6	0-1	0-1	0-1	0-1
	1201-3200	8	8	1-2	0-1	0-1	0-1
	3201-10000	10	10	1-2	0-1	0-1	0-1
	10001-35000	10	10	1-2	0-1	0-1	0-1

NOTES:
1/ Premature arming signal: defined as an arming signal earlier then specified in Dwg. 9292196.
2/ Premature firing signal: defined as a firing signal before minimum enable time specified in 9292196.
3/ Hazardous electronics assy: defined as an assy which fires after maximum set SD time by any mode.
4/ Early SD: defined as an assy which fires at or after enable time but before minimum set SD time.
5/ 50% of the sample shall be conditioned and functioned at -37°C (+0°, -3°) with remainder at 63°C (+3°, -0°).
6/ Handling procedure (see 4.5.18).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Electronics - MCD Assembly		9292225-1/-2
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Continuity of shorting bar across firing capacitor (6 to 9)		100%	3.5.5	4.5.16, 1/
Major					
101	Improper soldering of MCD flex cable		0.40%	3.2	Visual
102	Adhesive above MCD center cavity		0.40%	3.2	Visual
103	Improper assembly of MCD assy to electronic assembly (see Note 6 of Dwg)		0.40%	3.2	Visual
104	Resistance of receiver coil circuit (D,F)		100%	3.5.4	Ohmmeter, 4.5.15
105	Continuity of EBP series shorting bar (G,F)		100%	3.5.5	4.5.16, 1/
106	Resistance of fuse(s) circuit (10 to 9)		100%	3.2	Ohmmeter, 1/
107	Unactivated battery voltage check (8 to 3), (3 to 9)		100%	3.5.6	4.5.17, 2/
108	Incorrect lens identification, -1 versus -2		0.40%	3.2	Visual
109	Incorrect system lens, GATOR vs GEMSS, (9 - 12)		100%	3.2	Visual
110	AD switch missing, (P/N 9292225-2 only)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ Limit test current to one (1) ma. (See DWG 9292225).
2/ Use a voltmeter having a ten (10) megohm input impedance.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Bushing, Power Supply				9292172
					NEXT HIGHER ASSEMBLY 9292224
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	True position of I.D.		0.65%	3.2	Gage
203	Outside diameter		0.65%	3.2	Gage
204	Chamfer missing, one end		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Core, Toroid				9287094
					NEXT HIGHER ASSEMBLY 9292216
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Internal construction (see applicable note of dwg)	32	1/	3.2	Visual and microscope
102	Weld strength (see applicable note of dwg)	32	1/	3.2	Visual and manual
103	Constant-current flux reset test		100%	3.2	2/, Test equipment
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject lot if one (1) defective is found.
2/ See applicable note of drawing.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Magnetometer Assembly (before potting)				9287146
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of coil unwinding, or self bonding missing		0.40%	3.2	Visual
102	Conformal coating over entire winding and core		0.40%	3.2	Visual
103	Lead(s) improperly soldered to terminal(s)		0.40%	3.5.7	Visual, APPENDIX, 40.8
104	Resistance test		0.40%	3.2	Method 303, MIL-STD-202
105	Sensitivity test @ Z = 0 Oersted, 75° + 5°F	2/	0.40%	3.2	1/, Test equipment
106	Linearity test at Z = +0.45 Oersted & -45 Oersted, 75° + 5°F	2/	0.40%	3.2	1/, Test equipment
107	Sensitivity test @ temperature extremes, Z = 0 Oersted	2/	0.40%	3.2	1/, Test equipment
108	Impedance test		0.40%	3.2	1/, Test equipment
109	Solderability	20	3/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ See Dwg. 9287149.
2/ Majors 105, 106, 107 may be omitted if performed 100% on Electronic Assembly, Dwg. 9292223.
3/ Reject lot for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Capacitor, Fixed, Electrolytic-Tantalum, Solid				9287096
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current over max		0.40%	3.2	1/
103	Weld strength, two sides 1/	8	2/	3.2	Pull Tester
104	Internal workmanship	20	3/	3.2	DWG 9292220
105	Breakdown voltage	10	3/	3.2	DWG 9292220
106	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
107	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
108	Sleeve missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.
3/ Reject for two (2) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Capacitor, Fixed, Electrolytic-Tantalum, Solid				9287126
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current over max		0.40%	3.2	1/
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Weld strength (two sides), 1/	8	2/	3.2	Pull Tester
105	Internal workmanship	20	3/	3.2	DWG 9292220
106	Breakdown voltage	10	3/	3.2	DWG 9292220
107	Sleeve missing		0.40%	3.2	Visual
108	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.
3/ Reject for two (2) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Capacitor, Fixed, Plastic Dielectric (0.22 uF)				9287104
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance, 1/		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Capacitor, Fixed, Ceramic, Dielectric-Axial Lead				9292151 P/N 9292151-1/-2
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance, (leakage current)		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	1/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Capacitor, Ceramic, Fixed Dielectric, Molded, Axial	1 OF 1	9333929
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability, 2 leads	10	1/	3.5.7.1	MIL-STD-202, Method 208
105	Conformance to MIL-C-39014/5-2325			3.2	Certificate
106	Leakage current		0.65%	3.2	2/
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:
1/ Reject for one (1) or more defective.
2/ See applicable note of dwg.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Fuse, Electric		9328688
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridgewire material - free of etching compound			3.2	Certification
102	Fuse resistance		100%	3.2	Milliohmmeter, 3/
103	Case to lead resistance	1/	4/	3.2	Megohmmeter
104	No fire current for 10 second period, min.	1/,2/	4/	3.2	Tester
105	All fire energy (capacitor discharge) test	1/	4/	3.2	Tester
106	Solderability, 2 leads	10	4/	3.5.7.1	MIL-S-45743
Minor					
201	Fuse dimensioning	5	4/	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Sample size per MIL-STD-105, special inspection level S-4. 2/ Use these units for All Fire Energy Test (Major 105). 3/ Limit current to 1 ma. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 1/ RCR07G305JM/MIL-R-39008/1
4.4.2.16	Resistor (3 Megohms, ± 5%, 1/4 W)				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-39008/1
102	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ Applies to RCR07G104JM.
2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Resistor, Fixed Composition				RCR07G102JM/MIL-R-39008/1
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-39008/4
102	Solderability, 2 leads	10	1/	3.2	MIL-STD-202, Method 208
Minor					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287099-XXXX
4.4.2.18	Resistor, Fixed Film				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability, 2 leads	10	1/	3.2	MIL-STD-202, Method 208
Minor					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Resistor, Fixed Film		9287099-1611
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-39008
102	Solderability, 2 leads	10	1/	3.2	MIL-STD-202, Method 208
Minor					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ Reject for one (1) or more defective.

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Quality conformance inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Transistor (P/N 9287122-1, JAN 2N2907A)				9287122/MIL-S-19500/291
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire material (see applicable note of dwg)	32	1/	3.2	Visual
102	Wire bond (see applicable note of dwg)	32	1/	3.2	Pull Tester, 2/
103	Compliance with MIL-S-19500/291			3.2	3/
104	Solderability, 3 leads	10	1/	3.2	MIL-S-45743, 4/
105	Collector cut off current Icer over max.	6/	0.40%	3.2	MIL-STD-750, Method 3041 Condition B, 5/
106	Bond tails of internal interconnections incorrect (see applicable note of dwg.)	32	1/	3.2	MIL-STD-750, Method 2072
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective. 2/ Pull all wires on each device. 3/ Visual evidence of JAN marking. 4/ For gold plated leads MIL-S-45743 applies, otherwise MIL-STD-750, Method 2026 applies. 5/ See applicable note of dwg. for proper test time, voltage and load. 6/ MIL-STD-105, Level II, Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Transistor (P/N 9287122-2, JAN 2N2222A)	1 OF 1	9287122/MIL-S-19500/255
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire material (see applicable note of dwg)	32	1/	3.2	Visual
102	Wire bond (see applicable note of dwg)	32	1/	3.2	Pull Tester, 2/
103	Compliance with MIL-S-19500/255			3.2	3/
104	Solderability, 3 leads	10	1/	3.2	MIL-S-45743, 4/
105	Collector cut off current Icer over max.	6/	0.40%	3.2	MIL-STD-750, Method 3041, Condition B, 5/
106	Bond tails of internal interconnections incorrect (see applicable note of dwg.)	32	1/	3.7	MIL-STD-750, Method 2072
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective. 2/ Pull all wires on each device. 3/ Visual evidence of JAN marking. 4/ For gold plated leads MIL-S-45743 applies, otherwise MIL-STD-750, Method 2026 applies. 5/ See applicable note of dwg. for proper test time, voltage and load. 6/ MIL-STD-105, Level II, Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Network, Resistor-Fixed and Adjustable		9292219
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformance to MIL-N-63396		1/		
102	Solderability (all leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Incorrect identification/marking		1.0%	3.9	Visual

NOTES: 1/ Certification of conformance to MIL-N-63396 is required (see 6.2).
2/ Reject for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292137
4.4.2.23	Network, Capacitor-Resistor				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformance to MIL-N-63400		1/		
102	Solderability (all leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Incorrect identification/marking		1.0%	3.9	Visual

NOTES: 1/ Certification of conformance to MIL-N-63400 is required (see 6.2).
2/ Reject for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292138
4.4.2.24	Network, Capacitor-Resistance				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformance to MIL-N-63400		1/		
102	Solderability (all leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Incorrect identification/marking		1.0%	3.9	Visual

NOTES:
1/ Certification of conformance to MIL-N-63400 is required (see 6.2).
2/ Reject for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.25	Switch Assembly		9298512
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch function seven (7) contact closures minimum each direction	1/,2/	2.5%	3.2	4.5.28.1, 3/
102	Contact closure duration 0.60 seconds max.	1/,2/	2.5%	3.2	4.5.28.1, 3/
103	Insulation resistance	1/,2/	2.5%	3.2	4.5.28.1, 3/
104	Assembly damaged, distorted or incomplete		0.40%	3.2	Visual
105	Ribbon improperly welded to can		0.40%	3.2	Visual
106	Epoxy seal missing or incomplete		0.40%	3.2	Visual
107	Plating of AD switch can	5/ Batch		3.2	4.5.28.3
108	Solderability, 2 leads	10	0-1	3.5.7.1	MIL-S-45743
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

1/ All inspection lots of AD switches shall be subjected to 100% screening for proper function (Majors 101, 102 and 103) with all defectives removed. In addition to the screening operation, proper function shall be verified using the AQL's specified above.

2/ A defect is defined as: a switch with less than 7 momentary contact closures in each direction while being rotated 360° clockwise and counter clockwise; contact closure duration exceeding 0.60 second maximum; a switch with an insulation resistance, between switch terminals, of less than two (2) megohms. If any switch has any closure that exceeds 1.20 seconds the lot shall be rejected. The lot shall also be rejected if any switch has more than two contact closures between 0.60 and 1.20 seconds.

3/ See also drawing for test schematic and functional requirements.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Diode, Hot Carrier (JAN IN5711-MIL-S-19500/127)				9287169
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward voltage drop		0.40%	3.2	MIL-S-19500/127
102	Reverse leakage current over max.		0.40%	3.2	MIL-S-19500/127
103	Solderability (2 leads)	10	1/	3.2	MIL-STD-750, Method 2026
104	Evidence of JAN marking	10	1/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Semiconductor Device, Diode (Type JAN IN645/IN645-1, MIL-S-19500/240)				9287168
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Reverse leakage current over max.	1/	0.40%	3.2	MIL-STD-750, Method 4016dc
102	Forward voltage drop		0.40%	3.2	MIL-STD-750, Method 4011
103	Solderability (2 leads)	10	2/	3.2	MIL-STD-750, Method 2026
104	Evidence of JAN marking	10	2/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test, reverse current test is used to comfirm that the color band is on cathode end of diode. 2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.28	Semiconductor Device, Diode (Type JAN IN750A-1)		MIL-S-19500/127
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Internal visual inspection			3.2	Certification
102	Reverse leakage current over max-cathode band at wrong end or missing 1/		0.40%	3.2	MIL-STD-750, Method 4016dc
103	Forward voltage drop		0.40%	3.2	MIL-STD-750, Method 4011
104	Breakdown voltage		0.40%	3.2	MIL-STD-750, Method 4022
105	Solderability, 2 leads	10	2/	3.2	MIL-STD-750, Method 2026
106	Compliance with MIL-S-19500/127			3.2	Certification
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test, reverse current test is used to comfirm that the color band is on cathode end of diode. 2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.29	Amplifier, Operational, Triple Programmable Micropower		MIL-A-63242
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major				MIL-A-63242	
101	Wire bond strength	5	1/	3.1.7	Pull Tester
102	Wire material	.5	4/	3.1.7	Visual
103	Solderability	5	4/	3.1.4	MIL-STD-883, Method 2003
104	Electrical characteristics 3/			3.3.2	2/
105	Screening 3/			3.4	2/
106	External visual 3/			3.5	2/
107	Approved part/vendor verification	5	4/	3.1.7	Visual
Minor					
201	Incorrect identification/marking		0.65%	3.9 herein	Visual

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance to MIL-A-63242 required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Integrated Circuit, CMOS, Target Sensor		9287124
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire bond strength	5	1/	3.2	Pull Tester
102	Wire material	.5	4/	3.2	Visual
103	Solderability, all leads	5	4/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics 3/			3.2	2/
105	Screening 3/			3.2	2/
106	External visual 3/			3.2	2/
107	Approved part/vendor verification	5	4/	3.2	Visual
Minor					
201	Evidence of poor workmanship Item requires special handling - See 4.5.19.		0.65%	3.9 herein	Visual

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance to Dwg. 9287124 required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Microcircuit, Digital, CMOS (Initial Logic)	1 OF 1	9292962
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire bond strength	5	1/	3.2	Pull Tester
102	Wire material	.5	4/	3.2	Visual
103	Solderability	5	4/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics 3/		2/	3.2	MIL-M-63320
105	Screening 3/		2/	3.2	MIL-M-63320
106	External visual 3/		2/	3.2	MIL-M-63320
107	Approved part/vendor verification	5	4/	3.2	Visual
Minor					
201	Evidence of poor workmanship Item requires special handling - See 4.5.19.		0.65%	3.9 herein	Visual

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance to MIL-M-63320 required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292963
4.4.2.32	Microcircuit, Digital, CMOS (Final Logic)				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire bond strength	5	1/	3.2	Pull Tester
102	Wire material	.5	4/	3.2	Visual
103	Solderability	5	4/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics 3/		2/	3.2	MIL-M-63321
105	Screening 3/		2/	3.2	MIL-M-63321
106	External visual 3/		2/	3.2	MIL-M-63321
107	Approved part/vendor verification	5	4/	3.2	Visual
Minor					
201	Evidence of poor workmanship Item requires special handling - See 4.5.19.		0.65%	3.9 herein	Visual

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance to MIL-M-63321 required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Eyelet				9292146
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability	20	1/	3.2	MIL-STD-202, Method 208
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292191
4.4.2.34	Terminal, Stud				NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability	20	1/	3.2	MIL-STD-202, Method 208
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.35	Coil				9292232
					NEXT HIGHER ASSEMBLY
					9292244
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Inside first coil diameter, min.		0.40%	3.2	Gage
103	Thickness, max.		0.40%	3.2	Gage
Minor					
201	Length of flat, 2 places		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292230
4.4.2.36	Wiring, Printed, Flexible				NEXT HIGHER ASSEMBLY 9292244
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Largest outside diameter		0.40%	3.2	Gage
102	Total thickness, max.		0.40%	3.2	Gage
103	Diameter of "A" hole, 4 places		0.40%	3.2	Gage
104	Location of "A" hole, 4 places		0.40%	3.2	Gage
105	Diameter of "B" hole, 2 places		0.40%	3.2	Gage
106	Location of "B" hole, 2 places		0.40%	3.2	Gage
107	Profile incorrect		0.40%	3.2	Gage (Master template)
108	Conductor width and spacing incorrect		0.40%	3.2	MIL-P-50884
Minor					
201	Plating of pad incorrect 7 places (Note 3 of dwg)		0.65%	3.2	MIL-STD-275
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.37	Receiver Coil Assembly	1 OF 1	9292244
			NEXT HIGHER ASSEMBLY 9292231

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spacing of coil's last turn to edge of ckt board, min. - 3 places		0.40%	3.2	Gage
102	Incorrect soldering, 2 places		0.40%	3.5.7	Visual, MIL-S-45743
103	Thickness, max.		0.40%	3.2	Gage
104	Bonding of coil to flexible printed wiring incorrect		0.40%	3.2	Visual
105	Resistance		0.40%	3.2	Milliohmmeter
106	Inductance		0.40%	3.2	Impedance Bridge
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292231
4.4.2.38	MCD Assembly				NEXT HIGHER ASSEMBLY 9292225
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of poor soldering		0.40%	3.5.7	MIL-S-45743
Minor					
201	Protrusion of lead wires, max		0.65%	3.2	Gage
202	Wrong end of standoff adjacent to primer		0.65%	3.2	Visual
203	Height of primer above coil assembly		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292171
4.4.2.39	Standoff, EBP				NEXT HIGHER ASSEMBLY 9292231
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of EBP lead, 2 places		0.40%	3.2	Gage
102	True position of C.D. 101 Diam. with datum A		0.40%	3.2	Gage
Minor					
201	Width of groove		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.40	Primer, Electric Battery	1 OF 1	9292264
			NEXT HIGHER ASSEMBLY 9292231

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformance to MIL-P-63316		1/	3.2	
102	Solderability	20	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Certification of conformance to MIL-P-63316 is required (see 6.2).
2/ Reject for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9332372
4.4.2.41	Housing, Electronics Assembly				NEXT HIGHER ASSEMBLY 9292224
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Height to "D" hole flat (-B-)		0.40%	3.2	Gage
102	Radius of MDF cord groove, 2 places		0.40%	3.2	Gage
103	True position of outside diameter to datum "A"		0.40%	3.2	Gage
Minor					
201	Height to clearance flat (for cable)		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Spherical radius		0.65%	3.2	Gage
204	"D" hole diameter		0.65%	3.2	Gage
205	Overall width		0.65%	3.2	Gage
206	Width of test point hole		0.65%	3.2	Gage
207	Height of test point cavity		0.65%	3.2	Gage
208	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

* See 4.4.5 for inspection of mold features.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292243
4.4.2.42	Lid, Electronics Assembly				NEXT HIGHER ASSEMBLY 9292224
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Profile of MCD assembly nest		0.40%	3.2	Gage
102	Location of MCD assy nest with datum "A"		0.40%	3.2	Gage
103	Height of lid		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Inside diameter		0.65%	3.2	Gage
203	Diameter of terminal hole, 7 places		0.65%	3.2	Gage
204	Height of boss (for MCD) 2 places		0.65%	3.2	Gage
205	True position of outside diameter to datum "A"		0.65%	3.2	Gage
206	Diameter of EBP clearance hole		0.65%	3.2	Gage
207	True position of EBP clearance hole		0.65%	3.2	Gage
208	Thickness at base of MCD assy nest		0.65%	3.2	Gage
209	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

* See 4.4.5 for inspection of mold features.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.43	Connector, Edge Clip				9292156
					NEXT HIGHER ASSEMBLY
					9292158
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major 101	Solderability	20	1/	3.2	MIL-STD-202, Method 208
Minor	None defined				

NOTES:

1/ Reject for one or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292157
4.4.2.44	Shorting Bar, Ceramic				NEXT HIGHER ASSEMBLY 9292158
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness (substrate)		0.40%	3.2	Gage
102	Solderability	10	1/	3.2	2/
Minor					
201	Total height		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Dielectric glaze missing or incomplete		0.65%	3.2	Gage
204	Solder coating outside area "A" incorrect		0.65%	3.2	Visual
205	Thickness of Solder Pad (2 places)		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ Reject for one (1) or defective.
2/ Per MIL-STD-45743.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.45	Shorting Bar and Connector Assembly				9292158
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship incorrect		0.40%	3.5.7	Visual (10X)
102	Evidence of cracks or chips		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292197
4.4.2.46	Support Foam				NEXT HIGHER ASSEMBLY 9292224
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Adhesive missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.47	Power Supply Assembly				9292241
					NEXT HIGHER ASSEMBLY 9292223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformance with MIL-P-63421 (1/)			3.2	2/
102	Deformed dented or incorrectly oriented (polarity) battery cells		0.65%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Less Shorting Bar and Connector Assemblies (P/N 9292158) with associated components and Support Foam (9292197). 2/ Certification of Conformance to MIL-P-63421 is required (see 6.2).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63399B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.48	Electronics - MCD Assembly (Packaging, Marking, Packing, Shipping requirements)	1 OF 1	9292225-1/-2
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Nested foam liner damaged, 2 liners		0.40%	5.1	Visual
102	Conductive coat missing or improper, 2 liners		0.40%	5.1	Visual
103	Assemblies mixed, -1 with -2		0.40%	5.1	Visual
104	RF and static electricity protection gasket missing or defective		0.40%	5.2	Visual
105	Marking missing, incorrect or illegible		0.40%	5.3	Visual
106	Inspection lots not kept separate		0.40%	5.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection "Classification of Defects and Test" Forms, in 4.5 and as follows.

4.4.3.1 Ambient (25°C + 5°C), Hot (63°C + 3°C) and Cold (-37°C -3°C) temperature functioning tests. The table below shall be used for the selection of the lot acceptance sample sizes for ambient, hot and cold temperature testing of lots of electronic assemblies.

INSPECTION LOT SIZE	SAMPLE SIZE	* AMB	HOT	COLD	ACCEPT - REJECT CRITICAL/SPECIAL DEFECTS	MAJOR DEFECTS
2-500	32	12	10	10	0-1	0-1
501-3200	125	45	40	40	0-1	1-2
3201-10,000	200	68	66	66	0-1	2-3
10,001-35,000	315	105	105	105	0-1	3-4

The units shall be tested for all the functional parameters (under critical, special, major) listed in 4.4.2.4. Reject number is the number of defective units that would reject the lot. Since all the units have to be 100% inspected at ambient for critical and special defects the quantities listed under * do not have to be reinspected for critical and special defects at ambient temperature (see 4.5.24) but must be inspected for major defects.

4.4.3.2 Potting material testing. The potting material shall be checked for tensile strength, and hardness in accordance with DWG 9287148. Two test samples shall be molded per the drawing for each batch of mixed material or each 1,000 electronic assemblies assembled during continuous production, or each days production, whichever comes first. One test sample will be used for testing and one retained per the drawing. Identification shall be maintained between the sample and assemblies. Failure of the test specimen to comply with requirements shall result in rejection of the lot of electronic assemblies assembled from that batch of potting material. (See 4.5.27)

4.4.4 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the 'Paragraph Reference/Inspection Method' Column in Classification of Defects and Test Forms beginning with paragraph 4.4.2.1. The Contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6.3 herein.

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4.4.5 Dimensional control of molded parts. In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1), the contractor may, with the approval of the government, select a sample of three (3) parts (as molded) from each cavity to be fully inspected dimensionally (all dimensions on drawing) to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in pertaining paragraphs 4.4.2.1 through 4.4.2.48, after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the sample of three (3) parts, one (1) part shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part(s) will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part(s) shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAT-I for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAT-M. When a new cavity is substituted, an old cavity reworked, or the material or cycle time changed, a new curing time shall be established and approved (see 6.5).

NOTE 2: Those feature of a part which are subsequently machined or otherwise worked upon to meet its drawing requirements shall be inspected in accordance with the Classification of Defects.

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4.5 Test methods and procedures. Only government approved test equipment and test fixtures shall be connected to electronic assemblies under test to verify that it complies with the requirements specified herein. Increasing the PTB oscillator frequency to reduce test time is acceptable, provided the higher frequency oscillator signal maintains an approximate 50% duty cycle and has a peak amplitude frequency oscillator signal which does not cause or mask malfunctions in the functional operation of the electronic assembly; the higher frequency oscillator signal is well within the capability of the test equipment to monitor and control it with an accuracy of one cycle; the actual procedure used to increase the frequency of the signal is approved by the Government before it is implemented. When the PTB oscillator frequency is increased, the TTB oscillator frequency is to be proportionately increased or the TTB oscillator inhibited as applicable for the test being implemented. All testing shall be performed, unless otherwise specified at a supply voltage of 6.75 $\pm$ 0.15 volts, an ambient temperature of 25° $\pm$ 5°C, and a relative humidty of 60 $\pm$ 20%. Also, the rise time of the supply voltage applied to the electronic assembly under test shall not be less than 5 milliseconds, nor greater than 200 milliseconds.

4.5.1 Quiescent current test. With the electronic assembly under test connected to the test fixture and with the magnetic sensor, the piston actuator, and the detonator terminals each connected to the ground through a 10 ohm resistor, measure the quiescent current of the electronic assembly. The measured current shall be observed for compliance with the requirement specified in 3.5.2. This is a non-destructive test.

4.5.2 Primary Time Base (PTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base circuits oscillating at their natural frequency, measure the period of the PTB oscillator at test point TP4. Observe the measured period for compliance with the requirements specified in 3.5.3.1 for the corresponding temperatures and voltages under which this test is conducted. This is a non-destructive test.

4.5.3 Test Time Base (TTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base oscillators operating at their natural frequency, measure the period of the TTB oscillator at test point TP5. Observe the measured period for compliance with the requirements specified in 3.5.3.2 for the corresponding temperatures and voltage under which this test is conducted. This is a non-destructive test.

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4.5.4 Arming actuator signal test. With the electronic assembly under test connected to the test fixture, verify that the arming actuator signal occurs at the PTB clock count and temperature specified in 3.5.3.3. This is a non-destructive test.

4.5.5 Sensor enable test. With the electronic assembly under test connected to the test fixture, verify that sensor enable occurs at the PTB clock count, temperature and voltage specified in 3.5.3.4. Make the verification by incorporating, as a minimum, the following checks in the test procedure:

a. At the PTB clock count of 3 (±2) less than that required for sensor enable, activate magnetic sensor circuit and confirm that the detonator firing circuit does not turn on.

b. At the PTB clock of 3 (±2) greater than that required for sensor enable, activate magnetic sensor circuit and confirm that the detonator firing circuit is turned on.

This is a non-destructive test.

4.5.6 Low voltage detect test. With the electronic assembly under test connected to the test fixture, verify that the low voltage detect circuit initates the turn on of the detonator firing circuit at the temperature and voltage specified in 3.5.3.5. For this test it is required that both the PTB and TTB oscillators be operating at their natural frequency, and that the initial supply voltage be 6.75 ± 0.15 volts and linearly reduced at a rate of 0.40 (max) volts/second until the low voltage detect circuit is triggered. This is a non-destructive test.

4.5.7 Extended self-destruct signal test. With the electronic assembly under test connected to the test fixture and the fuses simulated open by connecting test point pad TP3 to the positive voltage supply, verify that the extended self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB clock count specified in 3.5.3.6. This test is to be conducted before the normal self-destruct test (4.5.8). This is a non-destructive test.

4.5.8 Normal self-destruct signal. With the electronic assembly under test connected to the test fixture, verify that the normal self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB clock count specified in 3.5.3.7. This is a non-destructive test.

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4.5.9 Magnetic sensor functioning test. The potted electronic assembly shall be connected to approved test equipment and tested for proper functioning of the magnetic sensor circuitry. The electronic assembly shall be tested at the magnetic latitude extremes specified on DWG 9292196 for the ambient all-fire and no-fire levels and firing angle specified on the drawing. High level cut-off functioning and spurious outputs shall also be checked at this assembly level. Observation shall be made for compliance with the applicable requirements of 3.5.3.8. Non-destructive test. NOTE: When testing for the magnetic function requirements specified in Table I of DWG 9292196, the acceptance test signal amplitude for values specified as " > " shall be between the value specified plus two (2) millioersteds, the acceptance test limit for values specified as " < " shall be between the value specified minus two (2) millioersteds.

4.5.10 Anti-disturbance function mode. With the electronic assembly under test connected to the test fixture so that its plane makes an approximate 45 degree angle with the horizontal, verify the anti-disturbance function mode by slowly rotating the electronic assembly about the cylindrical axis and confirming that the detonator firing circuit turns on before a 360 degree rotation is made. Observation shall be made for compliance with requirements 3.5.3.9. This is a non-destructive test.

4.5.11 PTB oscillator circuit check. With the electronic assembly under test connected to the test fixture and both the PTB and TTB oscillator circuits operating at their natural frequencies, verify that the oscillator check circuit is operational by confirming that the oscillator check circuit initiates the turn on of the detonator firing circuit within 32 TTB clock counts after the PTB oscillator is stopped. Observation shall be made for compliance with requirements of 3.5.3.10. This is a non-destructive test.

4.5.12 Firing circuit energy. With the elecronic assembly under test connected to the test fixture, verify that the piston actuator, Pin B-C, and parallel detonators firing circuits Pin A-C meet the applicable requirements of 3.5.3.11. This is a non-destructive test.

4.5.13 Self-sterilization function. With the electronic assembly under test connected to the test fixture and the detonator firing circuit an open circuit, verify that the voltage of the firing capacitor meets the requirements of 3.5.3.12. This is a non-destructive test.

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4.5.14 Reset circuit. With a linear increasing voltage (zero to 6.75 ± 0.15 d.c. volts in 50 milliseconds min.) applied to V_{DD} of the electronic assembly from an external power supply, verify that at reset the outputs at test point "B" and at test point "A" are logic low. Observation shall be made for compliance with requirements of 3.5.3.13. Non-destructive test.

4.5.15 Magnetic coupling device (MCD) assembly resistance. The countinuity-resistance of the MCD Coil assembled to the electronics assembly shall be measured using an approved ohmmeter (terminals F, D). Observation shall be made for compliance with requirements of 3.5.4. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.16 Shorting bar continuity. The continuity-resistance of each shorting bar circuit shall be measured, using an approved ohmmeter, between the test points 6 and 9 and between terminals "F" and "G". Observation shall be made for compliance with requirement of 3.5.5. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.17 Non-activated voltage. Each battery cell shall be checked for conformance with the requirements of 3.5.6 when measured by a 10 Megohm input impedance voltmeter. Non-destructive test.

4.5.18 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented and supplied to the Government (See 6.5). Verification of adherence to the cleaning procedure shall be done on a daily basis (Ref. Data Item DI-T-5372 (Tailored)).

4.5.19 Integrated circuit handling procedures. CMOS integrated circuits require care and handling to avoid damage due to static discharge. The contractor shall document his handling procedure for the CMOS integrated circuits and supply the procedure to the Government (See 6.5). Adherence to the handling procedure shall be verified daily. (Ref. DI-T-5372 (Tailored)).

4.5.20 Quality assurance soldering plan. The quality assurance soldering plan as noted in 3.5.7.2 shall be submitted to the Government (see 6.5). (Ref. DI-T-5372 (Tailored)).

4.5.21 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724. It shall include all detailed inspection and test results. Test data reports shall be submitted to the technical agency listed in 6.5.

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4.5.22 Variables data. The contractor shall submit variables data in accordance with Data Item Description DI-R-1724 for the first article samples and surveillance units. It shall include all detailed inspection and test results with respect to following parameters: Primary Time Base Oscillator Period, Test Time Base Oscillator Period, Quiescent Current, Low Voltage Detection Level, high and low Magnetic Sensor Threshold Levels. Test data reports shall be submitted to the technical agency listed in 6.5.

4.5.23 Surveillance units. The Electronics-MCD Assemblies, DWG 9292225-1/-2, randomly selected (see 3.8) for surveillance are to be stored at the contractors facility until contract completion and disposition made. One tenth of one percent (0.1%) or ten (10) units whichever is greater of each month's production shall be randomly selected. Ninety (90) percent of the units shall be 9292225-1 and remainder ten (10) percent shall be 9292225-2. The units are to be identified and serialized in a manner to provide traceability to year and week of manufacture with copy of data noted in 4.5.22 attached.

4.5.24 Ambient, hot and cold temperature functional testing. The sample of electronic assemblies specified in 4.4.3.1 shall be subjected to all functional tests at the operating temperature extremes. The units shall be conditioned at the specified temperatures (see 4.4.3.1) for two hours prior to the test. Testing shall be performed with the unit in the temperature chamber. Approved functional test equipment shall be used. Acceptable threshold levels are specified on DWG 9292196 for the temperature extremes. Acceptance limits for the digital circuitry (SD timing control) at the temperature extremes are specified on DWG 9292196.

4.5.25 Transportation - vibration. The lot acceptance test samples of electronics assemblies shall be serialized, packed as for normal shipment (see 5.0) and subjected to transportation vibration in accordance with Test 119, Procedure 2 of MIL-STD-331. Observation shall be made for compliance with requirement of 3.4.2. Samples shall be used in subsequent testing. First article only.

4.5.26 Self-Destruct (SD) mode destructive tests (Real Time). A randomly selected sample of Electronics - MCD Assemblies - P/N 9292225, with simulated resistive loads for the micropiston actuator and parallel detonators, shall have their batteries activated and allowed to function in the SD mode "Real Time" as follows:

a. Fifty (50) percent of the sample shall be placed in a fixture, providing similar confinement and resistive force of an S&A clamping plate to an ejected power supply bushing, then subjected to an electromagnetic pulse to activate batteries and set logic for normal SD. The remaining fifty (50) percent shall be set for extended SD.

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b. Functioning shall be at hot or cold temperatures.

c. Normal SD functioning shall be part of lot acceptance tests. Observation shall be made for compliance with requirement of DWG 9292196. Half of the units under test shall be conditioned and tested at 63°C (+3°, -0°) and the remaining half at minus 37°C (+3°, -0°).

d. Extended SD functioning shall be for 'Information Only' with test data supplied to DRDAR-QAT-M. Half of the units under test shall be conditioned and tested at 63°C (+3°, -0°) and the remaining half at minus 37°C (+3°, -0°).

e. Through use of fixtures and/or inclosures, the sample units shall be protected from radio frequency and magnetic field effect during the time interval covering activation of batteries through functioning.

f. If an electronic - MCD assembly fails to function by the maximum set self-destruct time, a valid target signature shall be applied (using a simulated prime target sensor signature coil). Any assembly failing to function (Non-hazardous dud) or that functions by sensor after set SD time (Hazardous dud) shall be set aside for dismantling and failure analysis.

g. Event times shall be monitored and recorded.

4.5.27 Potting material testing. Tests of the potting material shall be done in accordance with DWG 9287148 and 4.4.3.2 herein. Observation shall be made for compliance with requirements of 3.3.2 and 4.4.3.2.

4.5.28 Switch assembly (Anti-disturbance switch).

4.5.28.1 AD switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 360° on its cylindrical axis, while the axis is held at 15° ± 2° referenced to the horizontal, for both CW and CCW directions of rotation. Verify that the functional requirements of the drawing are met. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.28.2 Insulation resistance of AD switch. The switch assembly when in a suitable fixture shall be measured for insulation resistance in the reset position using an approved megohmmeter. Verify that the switch assembly does meet the requirements of the drawing. This is a non-destructive test and acceptable parts may be returned to the lot.

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4.5.28.3 Plating of AD switch can. Five (5) cans shall be randomly selected from each plating batch. The specimens shall be cross-sectioned through the axial center of the can, polished and examined microscopically to measure the plating thickness. If the total average plating thickness for the five cans fails to meet the requirements as specified on the drawing, the sample shall be classed defective and the lot shall be rejected. (Destructive test).

5. PACKAGING

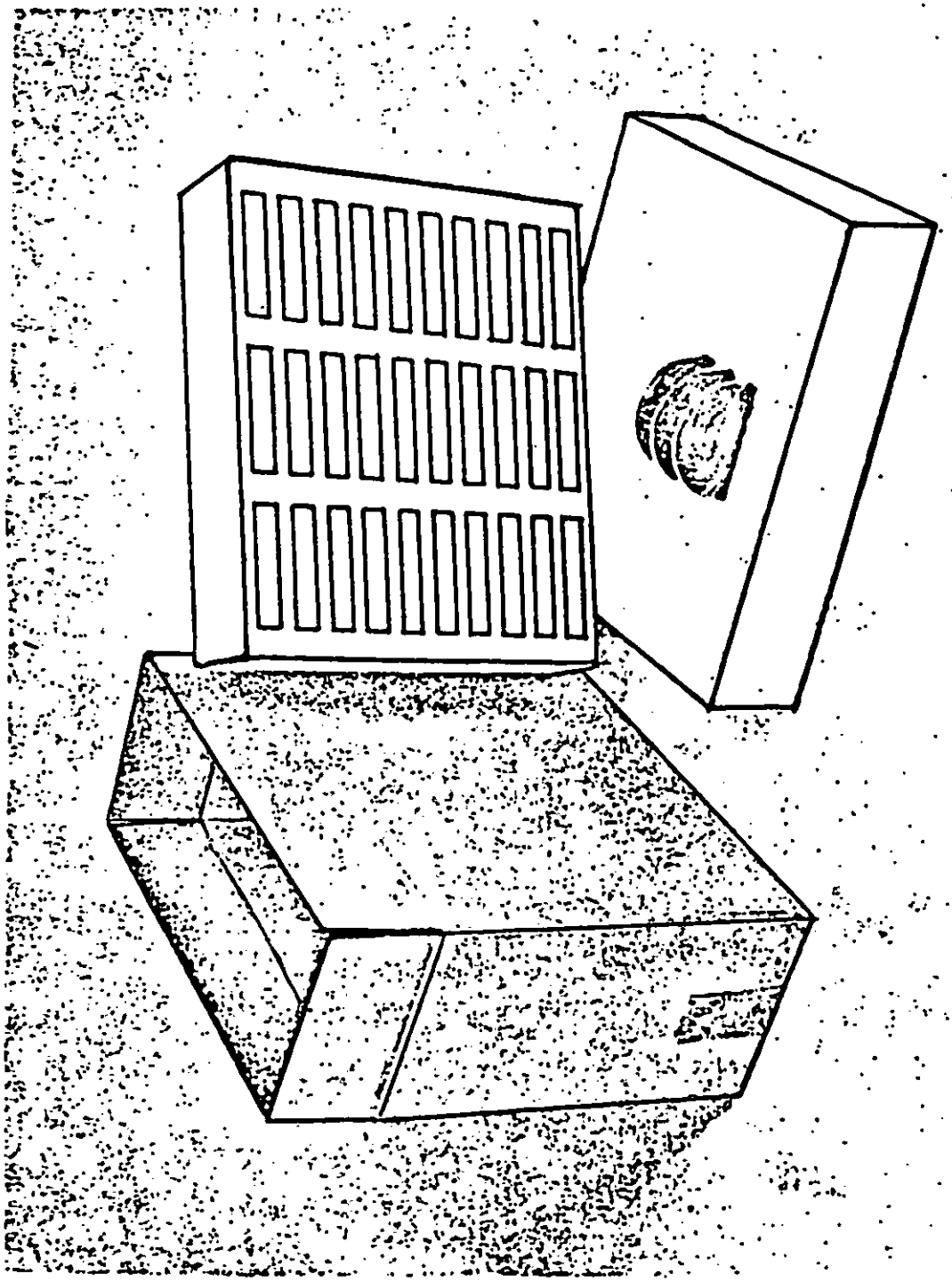
5.1 Packaging. The lenses shall be nest-packed in two identical nested foam liners which have been sprayed with a conductive coat. The liners assembly shall then be placed into a paperboard box. (See Figure 1).

5.2 Packing. The paperboard boxes noted in 5.1 above shall be inserted into M548 (Modified) ammo cans. To provide RF and static electricity protection, the M548 cans shall be modified to incorporate a carbon impregnated gasket. (See Figure 2 a, b).

5.3 Marking. The marking of the metal container shall be in accordance with MIL-STD-129 and DOT marking per code of Federal Regulations, Title 49, Parts 100-199. Fragile marking shall be used on the exterior of the packing.

5.4 Shipping. When electronics - MCD asemblies from more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

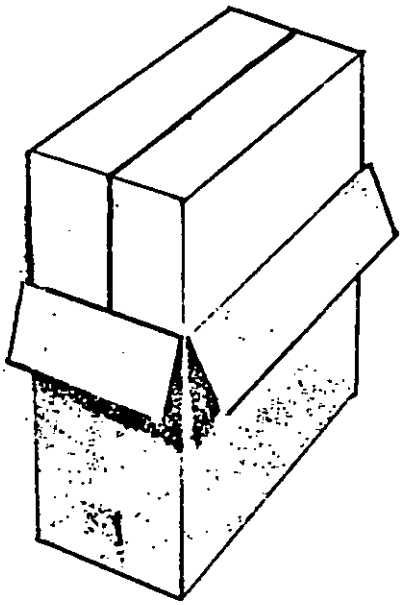
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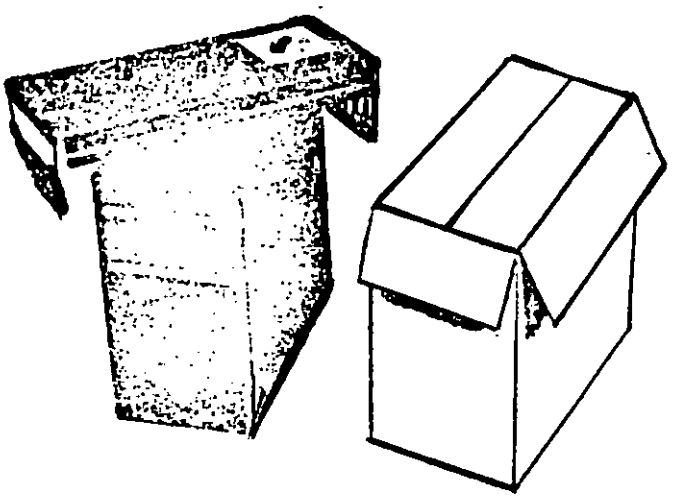
Electronics lens assemblies housed between identical foam liners

FIGURE 1. Packaging.

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a. Packing box concept for electronics lens assemblies



b. Packing box with M548 ammo can

FIGURE 2. Packing.

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6. NOTES

6.1 Intended use. The Electronics - MCD assembly covered by this specification is intended for use as a low power timing base and function generator for use in Mine, Antitank: HE, M75.

6.2 Ordering data. See MIL-A-48078. In addition the acquisition documents shall specify as required, the following:

a. Provisions for first article as specified in sections 3 and 4 of the following documents:

MIL-B-48761	-	Battery, Single Cell
MIL-P-63316	-	Primer, Battery, Electric Parts for and Loading, Assembling and Packing
MIL-M-63320	-	Microcircuit, Digital, CMOS, (Initial Logic)
MIL-M-63321	-	Microcircuit, Digital, CMOS, (Final Logic)
MIL-N-63396	-	Network, Resistor - Fixed and Adjustable
MIL-E-63399	-	Electroni - MCD Assembly
MIL-N-63400	-	Network, Capacitor - Resistor
MIL-P-63421	-	Power Supply Assemblies For Fascom Electronic Assemblies

b. Certification(s) of conformance for each lot or shipment (also see 6.6).

c. Provisions for surveillance data and units. (See 3.8, 4.5.23).

d. Process control documents (see 6.6).

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078) Submit equipment designs as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDDAR-QAT-I, Dover, New Jersey 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

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6.5 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, and process controls procedures shall include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.6 Data requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph	Data requirements	Applicable DID
6.2	Manufacturing Methods Report	DI-P-1604 (Modified)
4.5.18	Printed wiring board cleaning procedures	* DI-T-5372 (Tailored)
4.5.19	Integrated circuit handling procedure	* DI-T-5372 (Tailored)
4.5.20	Quality assurance soldering plan	* DI-T-5372 (Tailored)
4.4.2.4, 4.4.2.5	Quality Inspection Defect Report (Critical and Special Defects only)	DI-R-1721
3.6, 3.6.1 4.4.2.4, 4.4.2.5 4.5.21, 4.5.22 4.5.2.6	Quality Inspection Test Demonstration and Evaluation Report (Variable and real time data only)	DI-R-1724
4.4.2.22, 23, 24 29, 30, 31, 32, 40, 6.2	Quality Inspection Test, Demonstration and Evaluation Report	** DI-R-1724 (Tailored)
6.5	Ammunition Data Card	DI-L-1410

* Delete a, b, c, d of previous paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

** DI-R-1724 (Tailored) with block 10 modification dated 15 October 1981.

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6.7 Special defect. A defect, other than critical (see MIL-A-48078) that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

6.8 Change from previous issue. Due to the extensiveness of the changes, asterisks to identify changes with respect to the previous issue are not used in this revision.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1345-A164)

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APPENDIX

SUPPLEMENTAL SOLDERING PROVISIONS

10 SCOPE: This appendix details supplemental soldering provisions to MIL-S-45743 for hand soldering and to MIL-S-46844 for machine soldering.

20 APPLICABLE DOCUMENTS: See 2 herein.

30 EXCEPTIONS AND CLARIFICATIONS:

30.1 Exceptions/clarifications to MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraph titles shall apply.

Control of airborne contaminants. Delete the sentence, "Positive pressure shall ..... between them."

Soldering irons and resistance soldering irons. Delete the sentence, "The soldering iron shall.....or standby condition.", and substitute the sentence, "The soldering iron shall be a soldering iron approved by the technical agency or a soldering iron that attains a temperature no more than 10°F (5.6°C) above the control temperature when in an idling or standby condition."

Printed board terminations., Subparagraph 'b'. Change to read, ".... is permissible. Leads shall not extend, beyond the terminal pad or printed wiring pattern, far enough to cause less than .015 inches of clearance between conductors."

Printed board terminations., Subparagraph 'd'. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

Ribbon lead geometry. Add new sentence, "Ribbon lead termination from battery and lead assembly, shall comply with "Printed board terminations", subparagraph 'b'."

Cleaning the soldered connection. Change to read, ".... (within one (1) hour) using solvents or combination of solvents which will remove residual flux, activators, and other contaminants."

Resistivity of solvent extract. Delete.

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One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Sampling inspection. Delete.

Visual inspection. Change to read, ".... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

Cleanliness test. Delete.

30.2 Exceptions/clarifications to MIL-S-46844. The following exceptions/clarifications to reference MIL-S-46844 paragraph titles shall apply.

Flux application. Change to read ".... even coated surface. Any thinner used shall be a product from the flux manufacturer recommended for the specified flux being utilized or a substitute approved by the technical agency."

The solder contact. Add: "Bonded components or clinched leads are acceptable restraints provided they are implemented in a manner to assure that the leads do not move during the solder cycle."

Cleaning. Change to read ".... scraping shall not be used. Residual flux, activators and soils shall be removed to the degree approved by the technical agency using solvents or combination of solvents which will remove contaminants."

Resistivity of solvent extract. Delete.

One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Sampling inspection. Delete.

Visual inspection. Change to read, ".... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

Cleanliness test. Delete.

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Check the source to verify that this is the current version before use.

40 SUPPLEMENTAL SOLDER WORKMANSHIP STANDARDS

40.1 General. The following solder workmanship standards supplement the workmanship requirements contained in MIL-S-45743 and MIL-S-46844 and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspected connections may be examined using 10x magnification to determine acceptability.

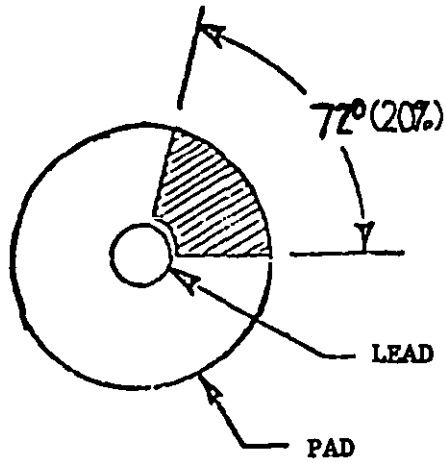
40.2 Definitions.

40.2.1 Voids. All holes, pits, and voids regardless of size are classified as voids.

40.2.1.2 Hole with solder plug. The total accumulative void area on each side of hole is less than 20% of the hole area and does not expose base metal. Also there are no voids through the board. Evidence of good wetting at the pad surface.

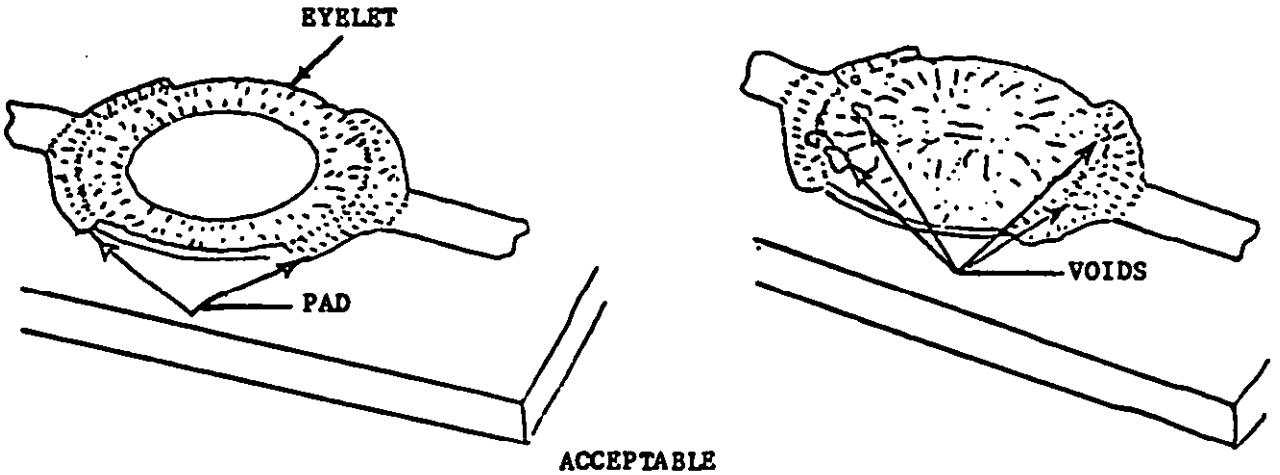
40.2.1.3 Hole with leadwire. For solder side and component side: the total accumulative void area is less than 20% of the lead circumference; does not visibly expose base metal; leads adjacent to voids exhibit good wetting and 100% solder coverage of the lead surface. There are no voids through the board. The lead and pad are wetted.

40.2.1.4 Twenty (20) % void. (For solder joint with a lead.) Total void area equivalent to an area subtended by an arc equal to 20% of the lead circumference.



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40.2.1.5 Eyelet connection voids. The total cumulative area of all voids is less than 20% of the total interface between eyelet and pad. Void(s) do not visibly expose base metal. The presence or lack of solder in the eyelet hole is not basis for nonacceptability.

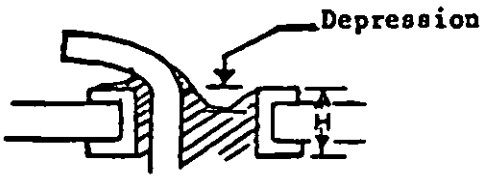


ACCEPTABLE

40.2.2 Maximum solder. A solder joint where it can be established that the lead exists and solder does not overflow beyond the confines of the connection area (the pad and tracks). Lead and pad must be well wetted.

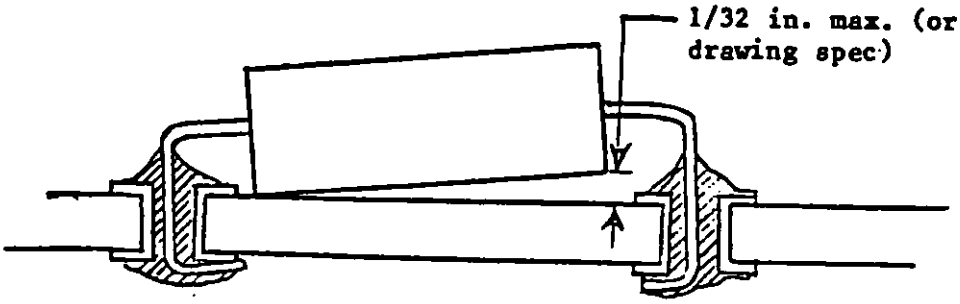
40.3 Lifting of pad. Lifted pads shall be controlled. Quantity of boards with lifted pads shall be controlled by the contractor as follows: Lifted pads exceeding 1/3 lifted to 0.40% AQL. Lifted pads less than 1/3 lifted is 1.0% AQL. Damage must not be in quadrant enclosing track. Damage not to break into plated through hole. No damage to opposite hole.

40.4 Lead, solder side. Exposed end (except copper) are visible at the end of the end. Lead is not closer than .015" to adjoining conductive circuitry. Maximum solder is permissible. Solder fillet is depressed, but the depression is less tna 25% of the board thickness.



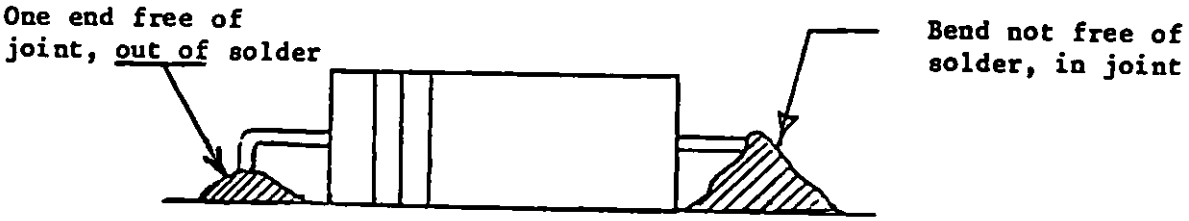
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40.5 Component alignment.

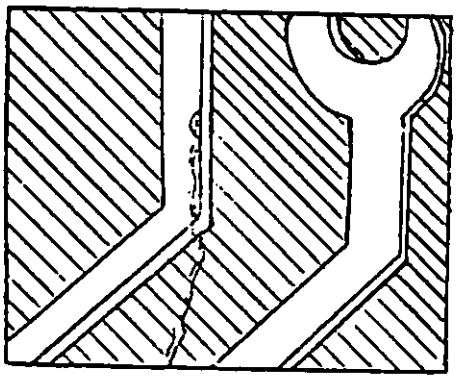


Vertical misalignment of component and printed wiring board is not in excess of 1/32 in. (or as specified on engineering drawing).

40.6 Strain relief. For axial lead components having a 0.080 max. diameter body allow one lead bend to be in solder joint, providing opposite lead bend is free of solder.



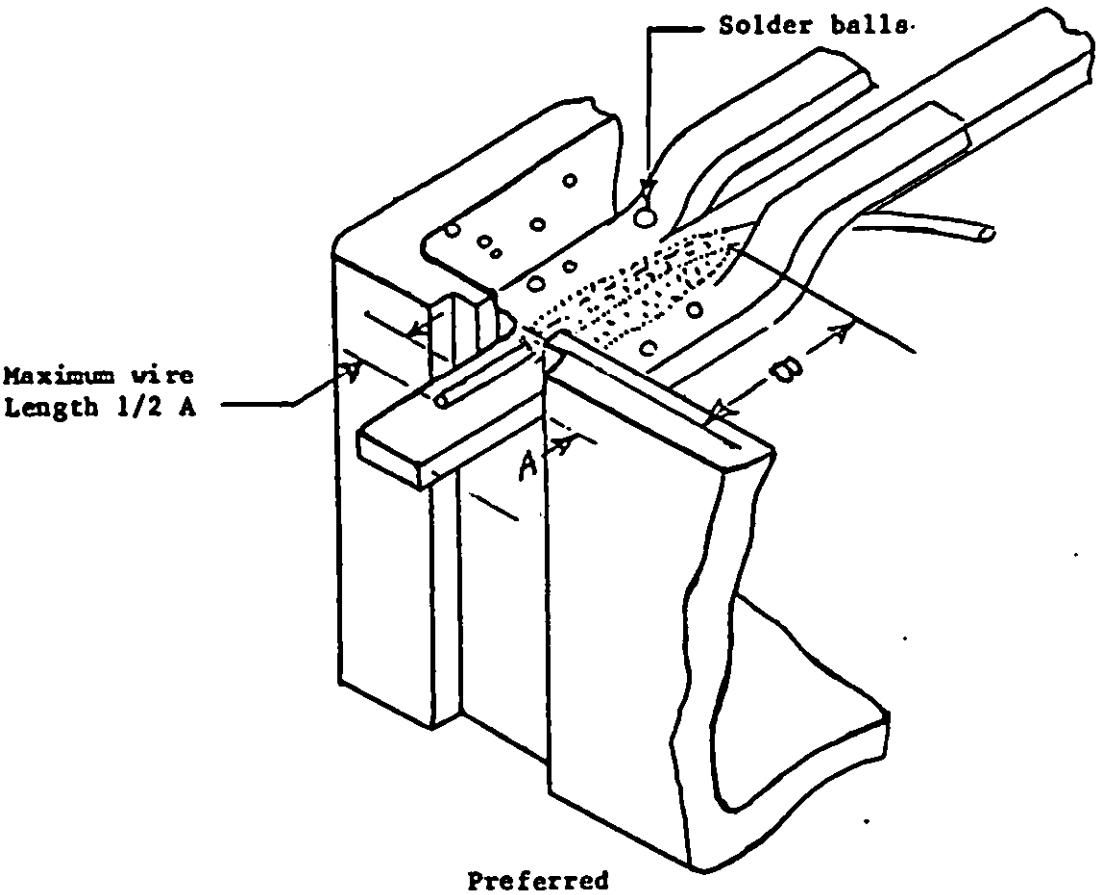
40.7 Scratches.



1. Scratch does not reduce cross-sectional area of conductor more than 20%
2. Does not visibly expose base metal.

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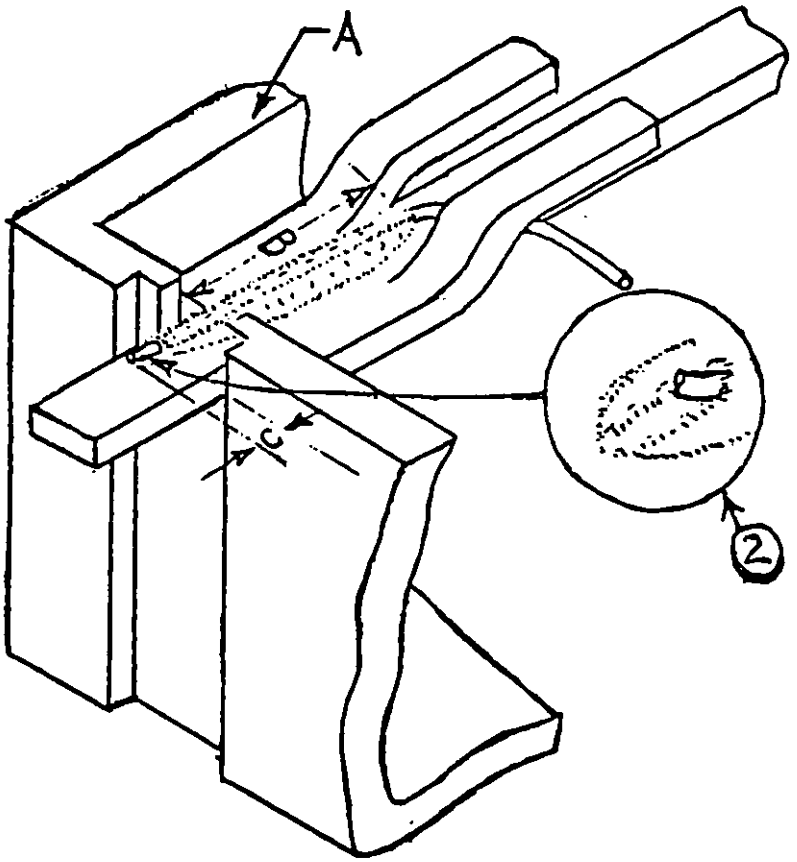
40.8 Magnetometer soldering. This section describes the acceptance criteria for solder connections of magnet wire to terminal. All examinations shall be performed with 10x magnifications (microscope).



Preferred

1. Good wetting of wire and terminal over entire length "B".
2. Solder is free of holes.
3. Contour of the wire visible.
4. Wire is extended along the length of terminal.
5. Protrusion of wire is less than one-half of Terminal length (A).
6. Solder balls are permissible in the potting area.
7. Bare copper end of wire is permissible.
8. Wire routed between terminal tabs, but not mechanical wrap is required.
9. Melted plastic may be present in the vicinity of solder joint (not allowed in solder joint).

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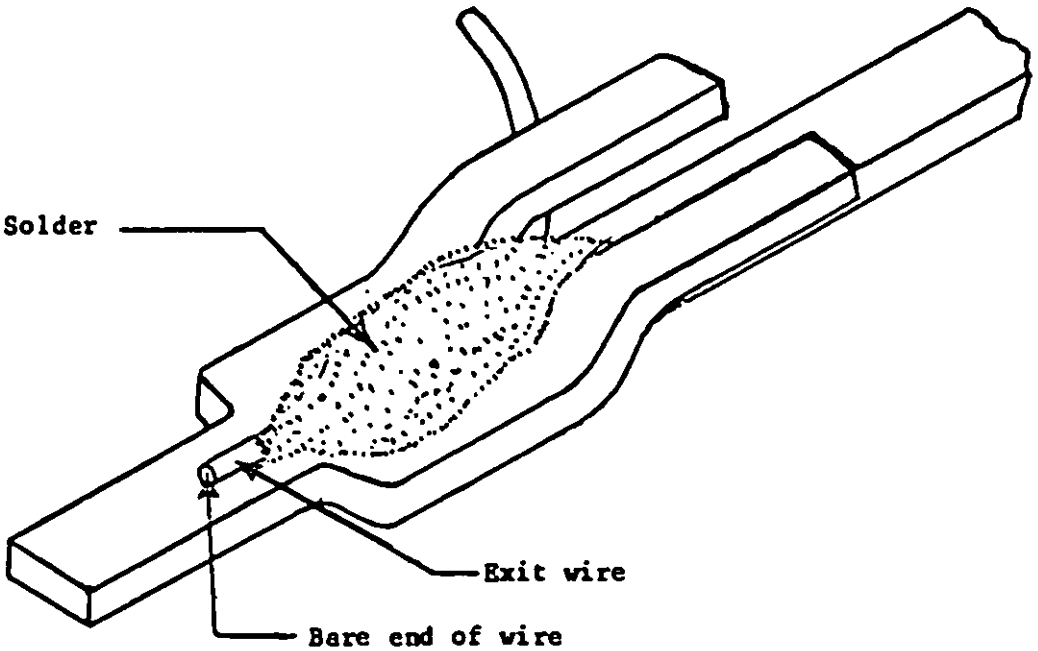
ACCEPTABLE PROTRUSION OF WIRE

1. Protrusion of the wire is above the surface of the terminal, but below the mold surface "A".

2. Imprint of wire after wire lifting during breakoff is permissible in area "C".

3. The lead must extend into area "C".

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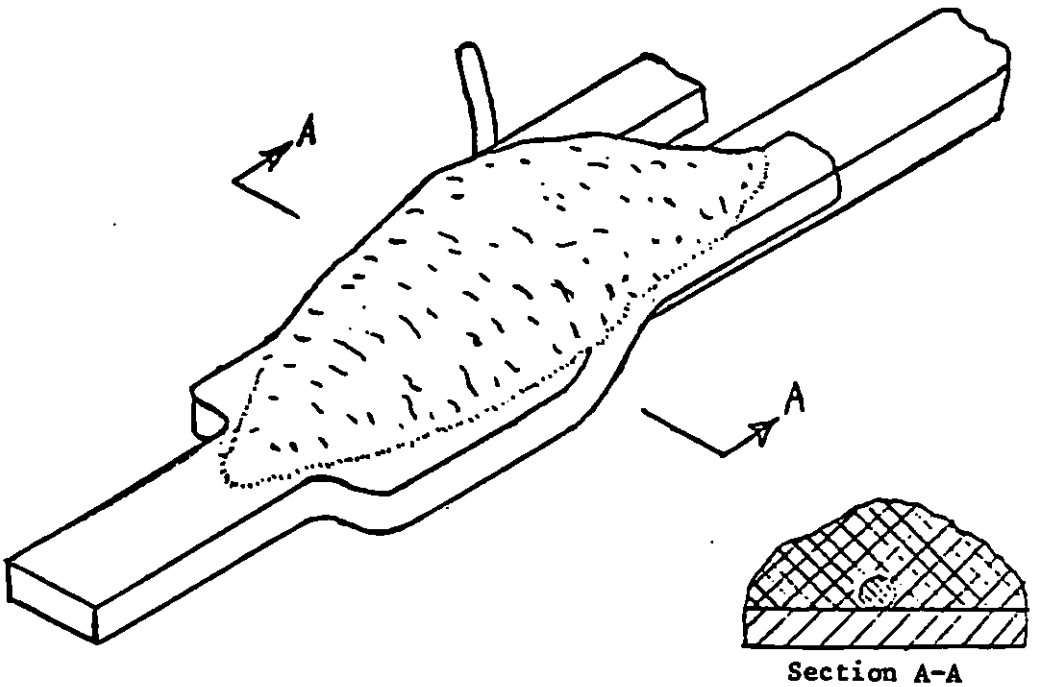
ACCEPTABLE

1. Bare end of wire is permissible.

2. Contour of wire not visible but wire ends are visible and show evidence of wetting entrance or exit.

3. Irregular surface; slightly convex fillet.

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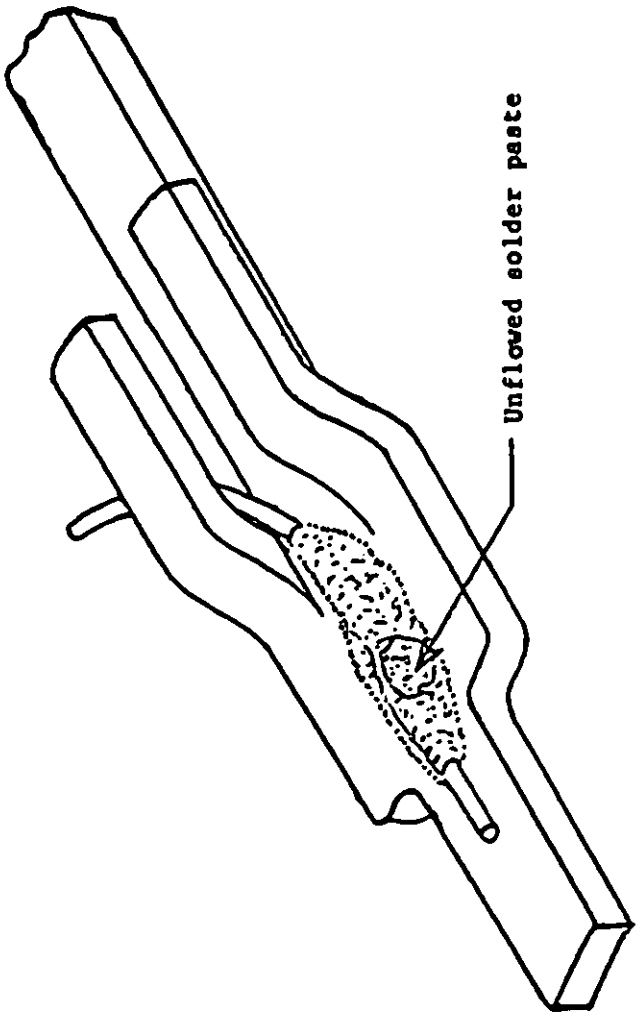
Section A-A

UNACCEPTABLE

1. The bare copper end of the wire is not visible.

2. Cannot be established that wire exists or that proper connection has been made.

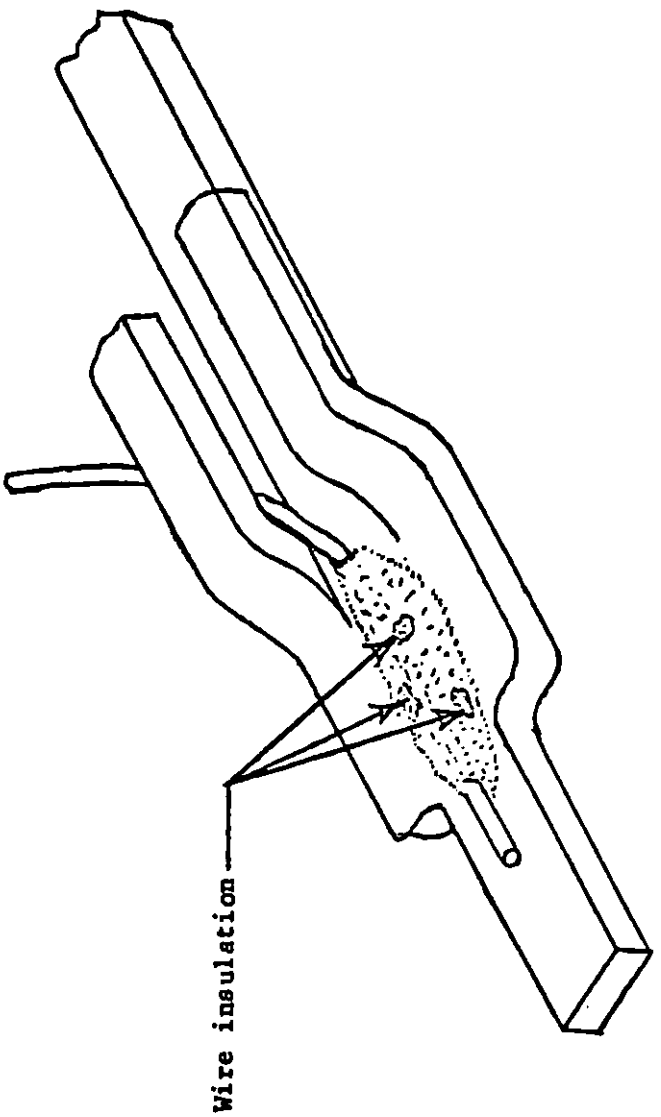
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UNACCEPTABLE

1. Evidence of solder paste in solder connection.

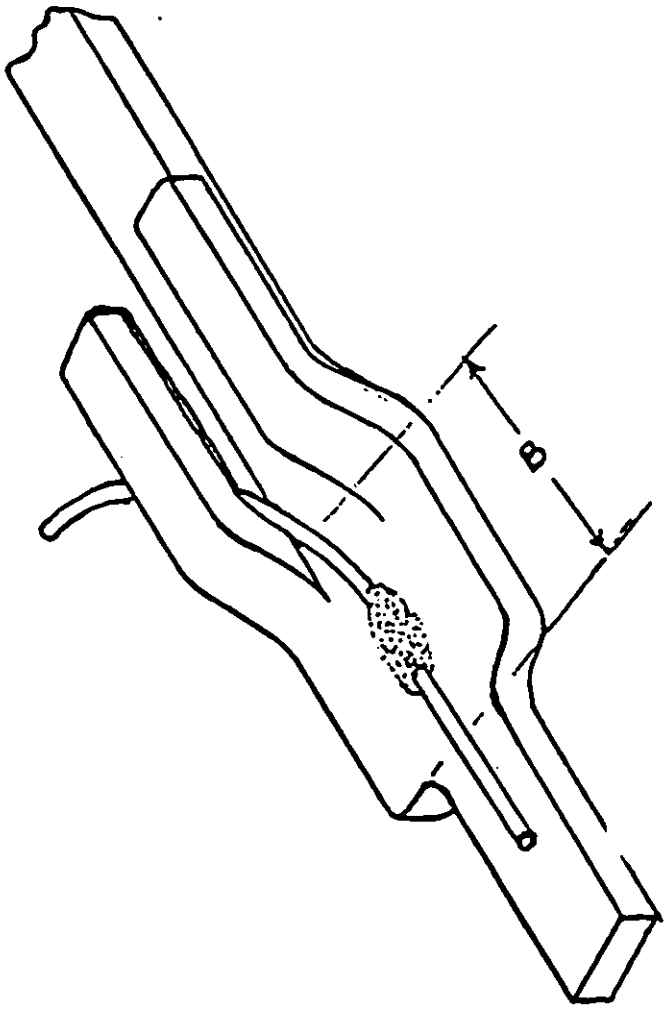
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UNACCEPTABLE

1. Evidence of contamination in solder connection.

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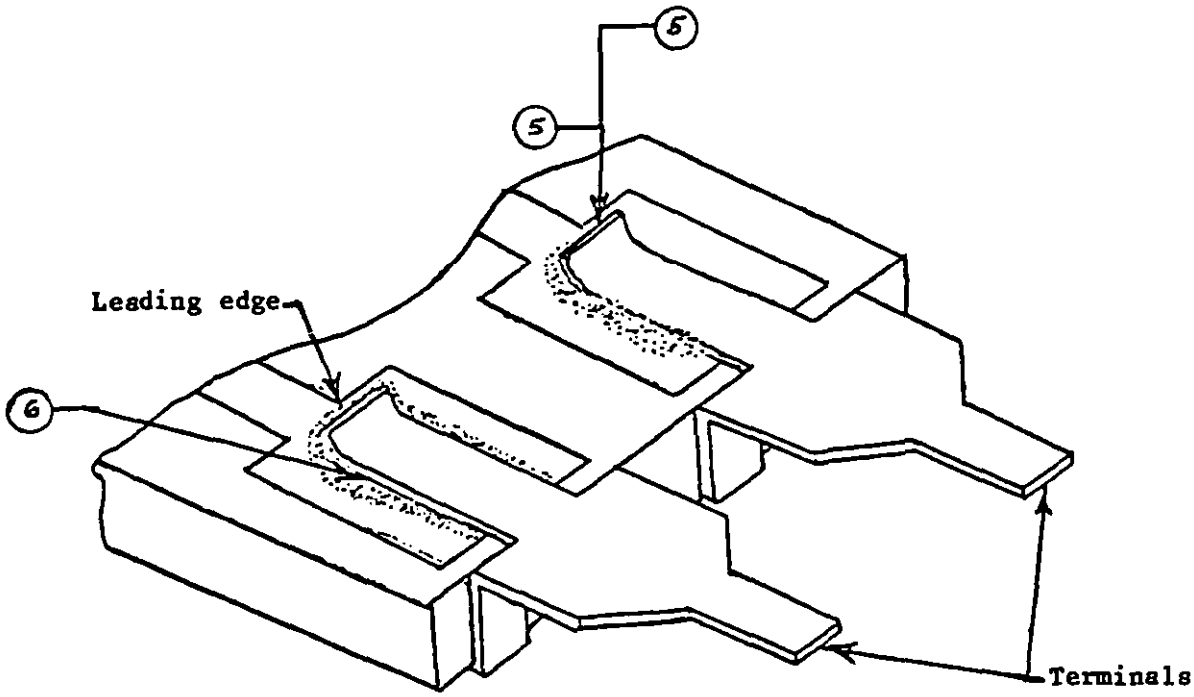


UNACCEPTABLE

1. Solder covers less than 50% of length "B".
2. Solder is not continuously wetted.
3. Voids exceeding 30% of solder joint present.

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40.9 Shorting Bar Soldering. This section describes the acceptance criteria for solder connections of terminal to pad on ceramic substrate. All examinations shall be performed with 10x magnification (Microscope).



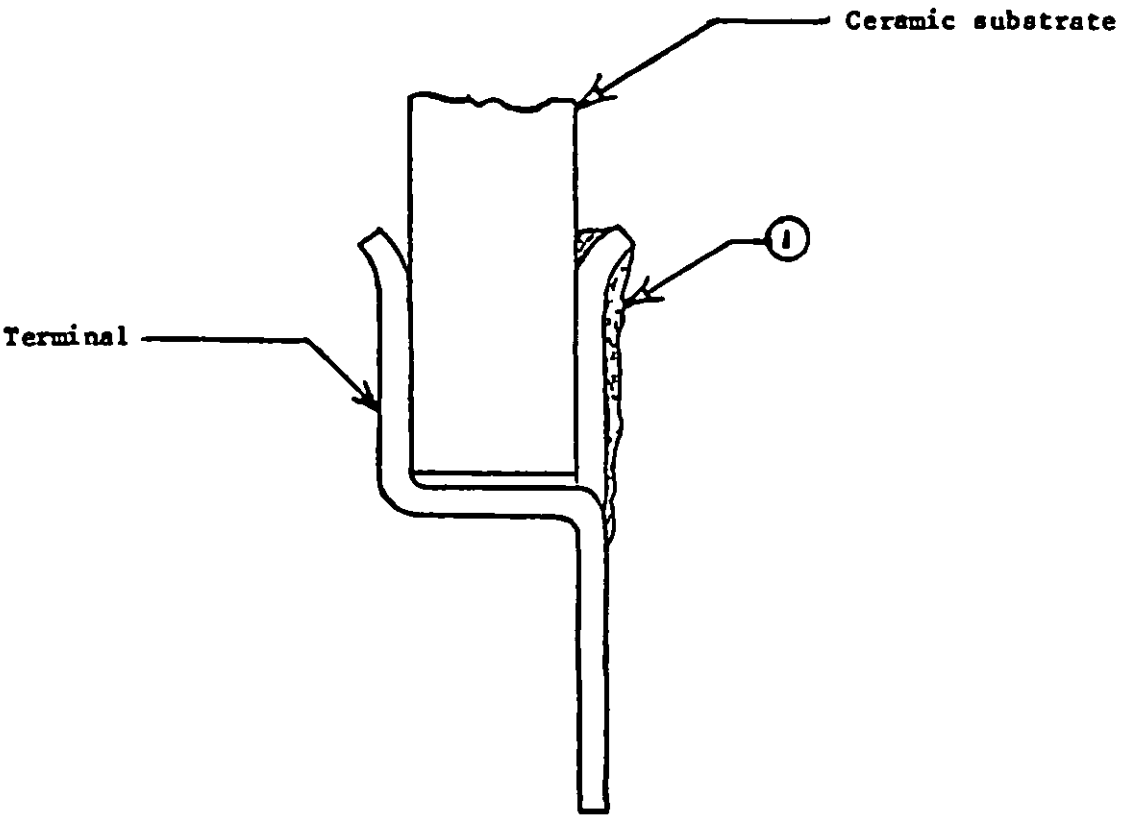
PREFERRED

1. Terminals are straight, parallel to and centered on solder pad.
2. Solder fillets are continuous on all edges for the entire length of terminal/pad interface without voids.
3. Solder wetting is evident on terminal edges and solder pad.

ACCEPTABLE

4. Solder is continuous and wetted on the entire leading edge and along both sides for 50% of the terminal/pad interface. No cracks or voids may appear between terminal and solder pad in the solder area.
5. Solder is continous and wetted on the leading edge and one (1) of two (2) edges of terminal for the entire length of the solder pad/terminal interface. No cracks or voids may appear between terminal and solder pad in solder area. There must be evidence of some solder wetting on opposite edge.
6. Solder fillet, 3/4 thickness of terminal, minimum.

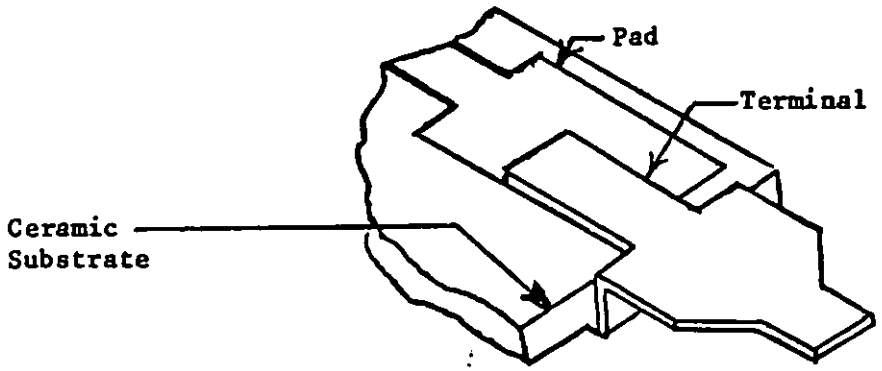
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ACCEPTABLE

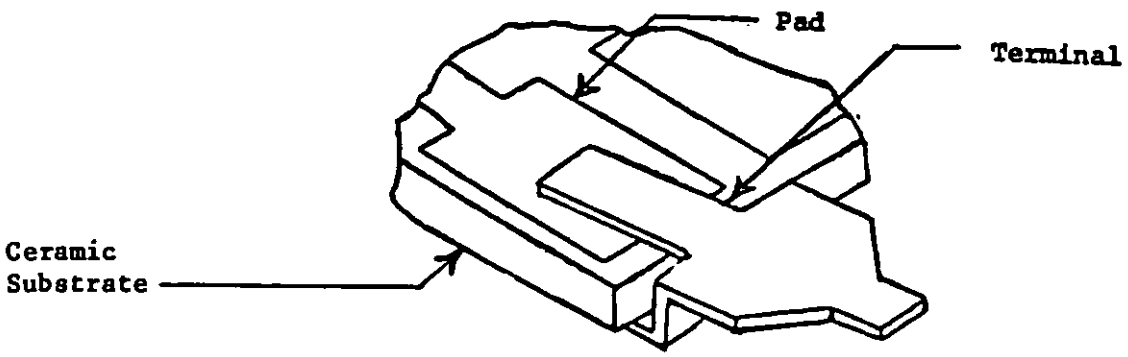
1. Solder buildup is not greater than one (1) terminal thickness.

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ACCEPTABLE MINIMUM

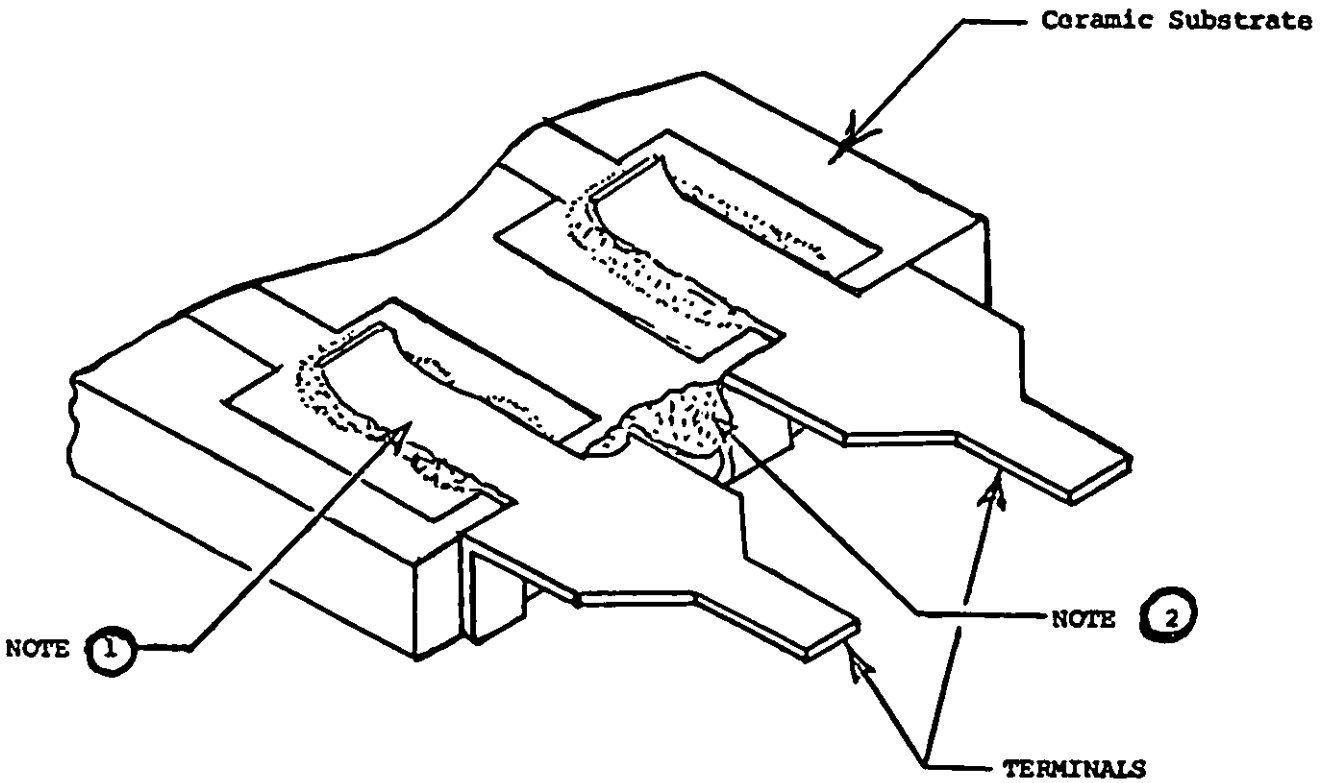
Terminal edge coincident with pad edge



ACCEPTABLE MINIMUM

Terminal slightly skewed, but does not extend beyond pad edge

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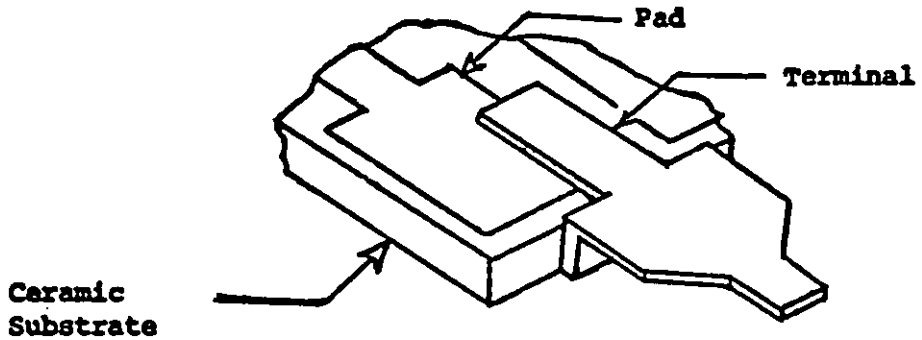
UNACCEPTABLE

NOTES:

1. Gaps in solder fillet exceed Note 4 of figure shown in paragraph 40.9.

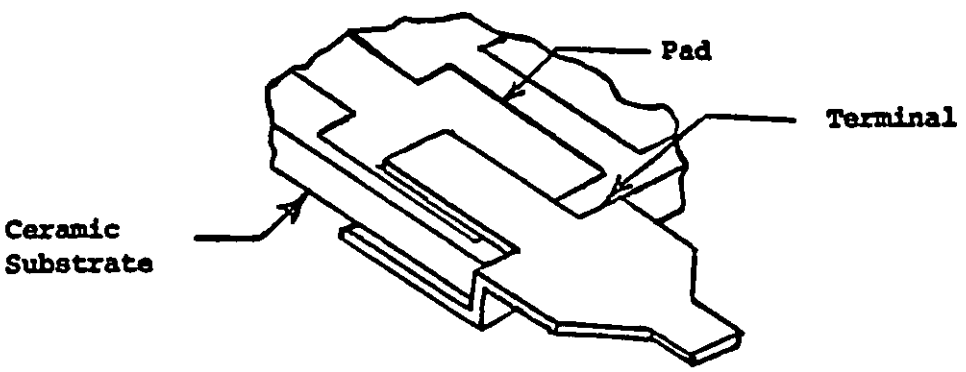
2. Bridging between terminals

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UNACCEPTABLE

TERMINAL OVERLAPS PAD



UNACCEPTABLE

Terminal overlaps ceramic substrate

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50 PRINTED WIRING BOARD ASSEMBLIES

Following wave soldering, printed wiring board assemblies shall be inspected (under adequate lighting and magnification) for soldering workmanship.

50.1 Major Defect Inspection

a. Inspection shall be to a 0.40 percent AQL, MIL-STD-105, Level II, normal sampling plan.

b. A board will be defined as defective during inspection if:

1. The bottom contains any joints defined as unacceptable by any of the documentation in sections 20, 30 and 40 of this appendix.

2. The top contains any joint defined as unacceptable by the documentation listed in 1 above except twelve (12) joints are allowed to be dewetted. A dewetted joint on the top must have evidence of solder in the hole when viewed from the top.

3. A maximum of thirty-five (35) joints with 0 to 20% voids will be allowed on each board; and no more than twenty (20) joints with 0-20% voids will be allowed on any one side of the printed wiring board.

50.2 Minor Defect Inspection

a. Inspection shall be on the sample size as specified in paragraph 50.1 above.

b. A board will be defined as defective if it contains three (3) or more minor defects.

c. Accept/reject criteria for the lot shall be to a 4.0% AQL, MIL-STD-105, Level II, normal sampling plan.

d. Minor defects are defined as the following:

1. Conductive surfaces closer than .010" but greater than .005" assembly characteristics only, not relative to the printed wire board construction.

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2. Copper showing -

(a) Clipped lead - evidence of good wetting and reflow must be present.

(b) Non-clipped lead - characteristic of good solder joint, i.e., wetting, etc., must be present.

3. Solder splash (attached and not shorted).

4. Solder balls less than .005".

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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